TECHNICAL RESEARCH REPORT — INTERVIEW PREPARATION

DC Fast Charger Controller, EV Communication Protocols, Charger-Side Hardware, and Buck/Boost Converters

A beginner-to-intermediate engineering guide covering: DC fast-charger architecture, charger-side and EV-side hardware, the complete EV↔charger communication flow (IEC 61851 control pilot, HomePlug Green PHY PLC, SLAC, DIN 70121, ISO 15118-2/-20), the charging sequence and state machine, buck and boost converter theory with worked examples, embedded control and firmware design, MATLAB/Simulink modelling, fault handling and troubleshooting, interview Q&A, practical diagnostic scenarios, and a study roadmap.

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How to read this report. Verified facts from public standards summaries and manufacturer documentation carry inline citations such as [TI SLVA057]. Statements that depend on a manufacturer, power rating, topology or protocol version are explicitly flagged *(architecture-dependent)*. Simplified illustrative examples are labelled as such and must not be treated as certified design data. The report is educational and is **not** a substitute for the actual standards (IEC 61851, ISO 15118, IEC 61851-23) or for your company's internal design documents. Selected difficult concepts include short Kannada explanations in parentheses: e.g., ಬಕ್ ಕನ್ವರ್ಟರ್ ವೋಲ್ಟೇಜನ್ನು ಕಡಿಮೆ ಮಾಡುತ್ತದೆ (ಇಳಿಸುವ ಪರಿವರ್ತಕ).

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1. Executive Summary

1.1 What is a DC fast charger?

A **DC fast charger (DCFC)**, also called a DC quick charger or — in standards language — an *EV Supply Equipment (EVSE)* providing DC charging, converts AC grid power into DC **outside** the vehicle and feeds that DC directly into the vehicle's high-voltage (HV) battery. This is the key contrast with ordinary AC charging: in AC charging the charger's onboard charger (OBC) inside the car performs the rectification, and its power rating (typically 3.7–22 kW *architecture-dependent*) limits the charging speed. In DC fast charging, the rectification and power conversion happen in the off-board cabinet, so power levels of 50 kW to 350 kW+ (and beyond, depending on the product) are possible, subject to what the vehicle's battery can accept [Heliox, DC Charging Hardware Guide — R7].

ಸರಳವಾಗಿ: AC ಚಾರ್ಜಿಂಗ್‌ನಲ್ಲಿ ಕಾರಿ ಒಳಗಿನ ಚಾರ್ಜರ್ ಹೆಚ್ಚಳ (ನಿಧಾನ), DC ಫಾಸ್ಟ್ ಚಾರ್ಜರ್‌ನಲ್ಲಿ ಬಾಹ್ಯ ಕ್ಯಾಬಿನೆಟ್ DC ನೇರವಾಗಿ ಬ್ಯಾಟರಿಗೆ ನೀಡುತ್ತದೆ (ವೇಗವಾಗಿ).

1.2 Why is a charger controller required?

The power electronics (rectifiers, DC/DC modules, contactors) can only move energy; they cannot decide *how much*, *when*, or *whether it is safe*. The charger controller — a microcontroller (MCU) or a cluster of MCUs plus a main controller — is required to:

- **Protect people and hardware**: run insulation (cable) checks before energising, verify connector lock, monitor over-current/over-voltage/over-temperature, and open contactors within milliseconds when a fault appears [Gongyuan IMD guide — R10].
- **Talk to the vehicle**: run the control-pilot signalling (IEC 61851-1) and the digital communication session (DIN 70121 / ISO 15118 over PLC) so that the charger delivers exactly the voltage/current the EV's battery management system (BMS) requests, and nothing more [Typhoon HIL ISO 15118-2 reference — R5].
- **Regulate the power stage**: continuously adjust PWM switching of the power modules through gate drivers, using closed-loop feedback from voltage and current sensors.
- **Manage the session**: meter energy, authenticate the user (RFID/app/Plug & Charge), talk to the charge-point operator's backend (OCPP), log events, and control cooling.

1.3 AC charging vs DC fast charging

Aspect	AC charging (Mode 3, Type 2)	DC fast charging (Mode 4, CCS2)
Where AC→DC conversion happens	Inside the vehicle (onboard charger, OBC)	In the off-board charger cabinet (rectifier + DC/DC stage)
Typical power	≈3.7–22 kW (limited by OBC) *(architecture-dependent)*	≈50–350 kW+ per dispenser *(product-dependent)*
What is on the "charging pins"	Live AC (L1/L2/L3), neutral	DC+ and DC− from the charger's power stage
Communication	Control Pilot PWM (IEC 61851-1) is sufficient; ISO 15118 optional	Control Pilot PWM **plus** digital PLC session (DIN 70121/ISO 15118) is required to negotiate DC voltage/current
Charger-side complexity	Low (relay, CP detection, metering)	High (PFC, DC/DC modules, liquid cooling, IMD, pre-charge, PLC modem)
DC output voltage range	Not applicable (no DC output)	e.g. ≈150–1000 V programmable output range in modern units *(varies by product)*

1.4 Power stage vs control/communication stage

A useful mental model for the interview: a DC fast charger is **two machines in one box**.

- **The power stage** — everything that carries the big energy: breaker, EMI filter, PFC rectifier, DC-link capacitors, isolated DC/DC converter modules, output contactors, connector pins, cooling. Rated in hundreds of kW.
- **The control/communication stage** — the MCUs, sensors, gate drivers, the CP circuit, the PLC modem (SECC), metering, HMI and backend connectivity. It senses, decides, commands, and reports. Rated in milliwatts, but it *governs* the power stage.

They meet at two places: **gate drivers** (control → power) and **sensors** (power → control). If you can explain this split clearly, you can explain the whole system.

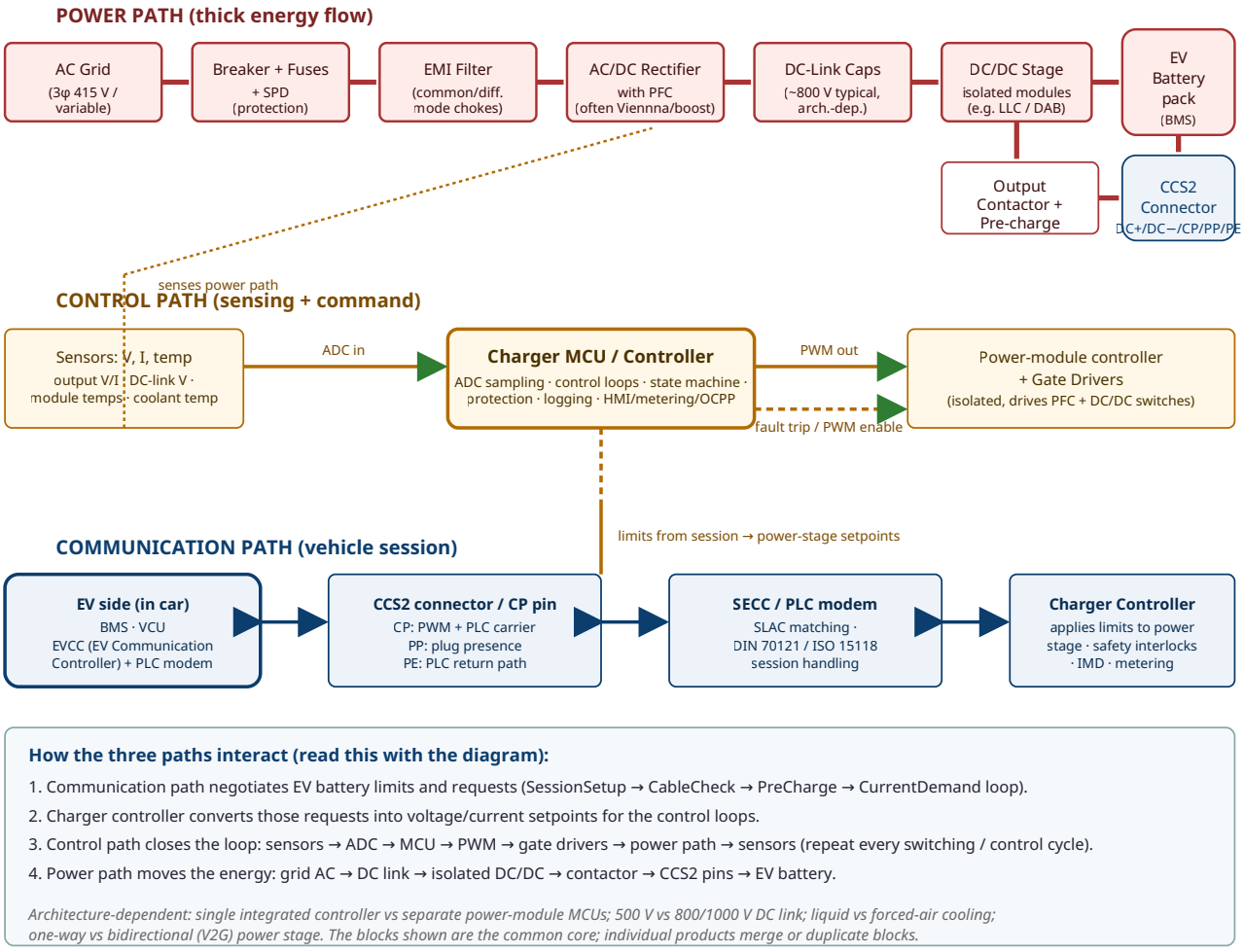
1.5 The complete system in simple language

You plug the CCS2 gun into the car. The charger detects the plug via the proximity pilot, hears the car "raise its hand" on the control pilot wire, and locks the connector. A small data modem then whispers over the same charging cable (power-line communication) to establish a digital session with the car's communication controller. The car says: "I am a 400 V battery, right now I want up to 250 A, and my maximum allowed voltage is 460 V." The charger checks the cable insulation with a test voltage, adjusts its output to match the battery voltage (pre-charge), closes the contactors, and starts delivering controlled DC power. Every second, the car keeps asking for the current it wants; the charger's feedback loops follow that request while watching voltage, current, temperature and insulation. When the battery is full or the car says stop, power ramps down, contactors open, the connector unlocks and the session ends. Everything else in this report is the engineering detail behind those two sentences.

2. Complete DC Fast Charger Architecture

2.1 Overall block diagram

Figure 1 — DC Fast Charger: power path, control path, communication path



How the three paths interact (read this with the diagram):

1. Communication path negotiates EV battery limits and requests (SessionSetup → CableCheck → PreCharge → CurrentDemand loop).
2. Charger controller converts those requests into voltage/current setpoints for the control loops.
3. Control path closes the loop: sensors → ADC → MCU → PWM → gate drivers → power path → sensors (repeat every switching / control cycle).
4. Power path moves the energy: grid AC → DC link → isolated DC/DC → contactor → CCS2 pins → EV battery.

Architecture-dependent: single integrated controller vs separate power-module MCUs; 500 V vs 800/1000 V DC link; liquid vs forced-air cooling; one-way vs bidirectional (V2G) power stage. The blocks shown are the common core; individual products merge or duplicate blocks.

Figure 1. Overall DC fast charger architecture. Red = power path, amber = control path, blue = communication path. All diagrams in this report are original.

2.2 Explanation of every block (power path)

Block	Purpose	Inputs	Outputs	Interaction with neighbours
AC input & distribution	Take 3-phase grid power, distribute to rectifier, auxiliary supplies and cooling	Grid (e.g. 400 V AC 3φ; India commonly 415 V; rating varies)	Protected AC feed	Feeds EMI filter; monitored by controller via input voltage sensing
Circuit breaker / fuses	Over-current and short-circuit protection, service disconnection	AC from grid	AC to charger (or nothing, if tripped)	Upstream of everything; controller can trip shunt-trip breaker on severe faults
Surge Protection Device (SPD)	Clamp lightning/grid surges	AC lines + PE	Clamped AC	Parallel device; its status flag can be read by the controller
EMI/EMC filter		Noisy AC	Cleaner AC	Between input and PFC; design must not

	Stop switching noise going to/from grid; also shapes the CP/PLC signal environment			attenuate the PLC band (≈2–30 MHz) [ISO 15118-3 summary — R4]
AC/DC rectifier with PFC	Convert AC to DC with sinusoidal grid current (power factor ≈1); boost the bus to a regulated DC-link voltage	AC mains	DC-link voltage (e.g. ≈800 V, *varies*)	Switches driven by gate drivers; PFC control loop in module controller
DC-link capacitors	Energy buffer; absorb switching ripple; hold voltage stiff between PFC and DC/DC	Rectifier DC	Smooth DC bus	Must be pre-charged at start-up and discharged at shutdown; monitored for over-voltage
DC/DC power conversion stage	Isolate and regulate the EV battery voltage/current from the fixed DC link (galvanic isolation required in DC charging per IEC 61851-23 practice)	DC link	Programmable DC output (V set by EV request)	The heart of the charger; often multiple parallel/ interleaved modules; controlled by MCU PWM via gate drivers
Output contactor + pre-charge/ discharge	Connect/disconnect output safely; pre-charge matches output voltage to battery before closure; discharge bleeds output after session	DC/DC output	DC to connector	Commanded by controller; contactor feedback (aux contacts) monitored
CCS2 connector + cable	Physical interface: DC+, DC−, CP, PP, PE	Contactor output	Energy + signalling to EV inlet	CP line carries both PWM and PLC; PP used for plug detection and (in AC) cable current rating

2.3 Control and communication blocks

Block	Purpose	Inputs	Outputs / interactions
Sensors (V/I/T)	Measure output voltage, output current, module/ component temperatures, DC-link voltage; provide the raw feedback for all control and protection	Power-stage nodes	Analog (or digital) signals to MCU ADCs / isolated digital interfaces
Charger MCU / controller	Runs state machine, control loops, protection, communication stack handling, logging, HMI/ metering/backend tasks (often split across several processors)	Sensor data, CP state, PLC session data, user input	PWM to gate drivers, contactor commands, cooling commands, displays, OCPP messages
Gate drivers / module controller	Translate MCU PWM into isolated, high-current gate pulses for power switches; provide dead-time, desaturation/over-current protection per switch	PWM + enable from MCU, DC-link supplies	Gate pulses to PFC and DC/DC switches; fault flags back to MCU
IMD	Insulation Monitoring Device: measures insulation resistance between DC output and PE; mandatory for the cable check and continuous monitoring in DC charging	DC output nodes, PE	Insulation status to controller and (via protocol) to the EV
CP / PP interface	Generate and sense the control-pilot voltage/PWM; sense proximity/presence	CP line, PP line	Vehicle state (A/B/C/D/ E/F) to MCU
SECC (PLC modem)	Supply-Equipment Communication Controller: SLAC pairing and the DIN 70121 / ISO 15118 session over HomePlug Green PHY PLC	PLC carrier on CP+PE	Session messages to controller; requested limits to control loops
Auxiliary power supply	Generates low-voltage rails (e.g. 24 V, 12 V, 5 V, 3.3 V) for MCUs, drivers, relays, cooling	AC input or DC link	Standby power — present even before a session starts

2.4 Architecture-dependent variations (say this in the interview)

- **Topology of the power stage**: many commercial chargers use a two-stage design (PFC rectifier → isolated DC/DC such as LLC or dual-active-bridge). Some modular designs stack or parallel modules; some use interleaved phases to reduce ripple. Not every charger is "a buck or a boost" — see §9.4.
- **Control partitioning**: a single main controller may run everything, or the power modules may have their own controllers that only accept setpoints from the main controller. Both exist.
- **Voltage class**: 150–500 V, 150–850 V, 150–1000 V output windows are common product classes; newer products extend to 1000 V+ for 800 V-class vehicles *(product-dependent)*.
- **Cooling**: forced-air below ≈60 kW class, liquid cooling for high power modules and cables *(typical practice, varies)*.

3. Complete Charger-Side Hardware and Components

Key for the table below: Essential = present in essentially every DC fast charger; Arch-dep. = depends on design/power class/region; Optional = market or feature dependent. Ratings shown are *typical illustrative ranges* — always verify against the specific product datasheet; do not quote them as standard values in an interview without saying "typical".

Component	Main function	Input → Output	Typical signals / ratings (illustrative)	Connection to MCU	Common failure modes	Test / debug method	Class
AC input terminal & distribution	Terminate grid feed; distribute to power stage, aux PSU, cooling	Grid 3φ → internal busbars	e.g. 400/415 V 3φ, 50/60 Hz; up to hundreds of A per phase	Input V/phase presence sensing; breaker status aux contact	Loose termination → arcing/heating; phase loss	Thermal camera on terminations; phase-presence indicators; torque checks	Essential
Circuit breaker(s) & fuses	Overload/short-circuit protection; isolation for service	Grid AC → protected AC	MCCB sized to feed; semiconductor fuses inside power modules	Aux contacts → digital inputs	Nuisance trips; welded/fatigued contacts; wrong sizing after retrofits	Trip counters in logs; contact-resistance test; verify coordination with fuse curves	Essential
SPD (surge protector)	Clamp lightning/switching surges	AC+PE → clamped	Type per region (e.g. T1/T2); MOV-based	Digital status (healthy/end-of-life flag)	MOV degradation after surges; silent end-of-life	Read status flag; visual window check during service	Essential
EMI/EMC filter	Suppress conducted EMI both directions; must pass PLC band on CP/PE	Noisy AC/DC lines → filtered	CM/DM chokes, X/Y caps	None (passive)	Capacitor failure; over-temperature; PLC signal attenuation if mistuned	Conducted-emission pre-compliance scan; PLC attenuation test (SLAC margin)	Essential
AC/DC rectifier (front end)	AC→DC conversion	AC → DC link	Diode-bridge (small) or active 3φ PWM rectifier (most DCFCs)	Module controller / gate drivers	Device failure (short), driver failure, DC-link cap failure	Oscilloscope on DC link; device curve-tracing in service; logs of PFC errors	Essential
PFC stage	Force grid current sinusoidal (PF≈1); regulate DC-link voltage	AC → regulated DC link	Boost/Vienna-type; PF>0.98 typical target; THDi limits per grid codes	PFC controller (often inside module controller); PWM from MCU-level logic	Loss of PF correction → grid-code failures; unstable DC link	Power analyzer (PF, THD); DC-link ripple measurement	Essential for DCFC
DC-link capacitors	Buffer energy, stiffen bus	Rectifier → smooth DC	Electrolytic or film caps; e.g. bus ~800 V class *(varies)*	Voltage sense; temp sense (arch-dep.)	Drying out (electrolytic); swelling; venting; capacity fade → higher ripple	Capacitance/ESR measurement at service; ripple spectrum on scope	Essential
DC/DC converter /	Isolated conversion DC link → battery		Full/half-bridge LLC, DAB, phase-shift full	Module controller accepting V/I	Switch short-circuit, transformer	Setpoint step-response tests; module error	Essential

power modules		DC link → controlled DC out	bridge; modular stacks	setpoints; status CAN/ UART to main MCU	insulation, control drift	registers; thermal scan	
MOSFET / IGBT / SiC MOSFET switches	High-speed switching of power stage	—	IGBT 1200 V class (older), SiC MOSFET 1200 V class (modern, higher efficiency & frequency) [Infineon CoolSiC EV charging portfolio — R13]	Driven via gate drivers	Overcurrent/ avalanche failure, gate-oxide damage, solder fatigue	Desat trip tests; V_{CE}/$R_{DS(on)}$ trending; IR thermography	Essential
Gate drivers	Isolate & amplify PWM; protect switches	Logic PWM → isolated gate pulses	Isolated drivers; ±V gate rails; desat/OC detection	PWM in, fault line out	Bootstrap cap failure, opto/ isolation degradation, false turn-on	Scope on gate waveform (V_{GE}/V_{GS}, timing, ringing); fault-flag injection	Essential [TI SLUA618 — R12]
Voltage sensors	Measure DC-link & output voltage	HV node → scaled signal	Isolated (Hall/ AMR) or resistive dividers with isolation amps	ADC input (isolated)	Gain/drift error; damaged isolation	Calibration against reference meter; plausibility checks in firmware	Essential
Current sensors	Measure output current(s) & per-phase currents	Conductor → signal	Shunt (+isolated amp), Hall-effect (closed/ open loop), fluxgate for precision	ADC input	Offset drift, saturation (Hall), thermal shift	Clamp-meter cross-check; offset calibration at zero current	Essential
Temperature sensors	Module/inductor/ coolant/connector temperatures	—	NTC thermistors, PT100/RTD, on-die sensors in smart modules	ADC / digital (I2C/SPI)	Open/short reading as extreme values; detachment → wrong temp	Readback plausibility; compare channels; IR camera spot check	Essential
MCU / embedded controller	Brain: state machine, loops, comms, protection	Sensors, comms → commands	32-bit MCUs (ARM Cortex-M/R class); sometimes multiple MCUs + a main CPU	— (is the hub)	Watchdog resets, flash corruption, brown-out	Debug logs, watchdog-reset counters, JTAG/SWD traces in development	Essential
ADC & PWM peripherals	Digitize sensors; generate switching/ control PWM	Analog in → digital values; timer → PWM	12–16-bit ADCs; PWM ~kHz (low-voltage rails) to tens of kHz–100 kHz+ per module *(varies)*	On-chip	Timing jitter, sampling mis-synchronization	PWM scope checks; ADC self-test routines	Essential
Auxiliary power supply	Low-voltage rails for everything electronic	AC/DC link → 24/12/5/3.3 V	Standby operation always on	Powers MCU, drivers, relays, comms	Caps aging, brown-out during grid dips	Rail voltage checks under load; brown-out ride-through tests	Essential
Contactors & drivers	Connect/disconnect DC output	DC output path	DC-rated contactors (magnetic blowout /	Digital output + aux-contact feedback input	Contact welding, coil burnout, slow operation	Aux-contact timing measurement; welding detection	Essential

			COSEN type); coil 24 V typical			routine at session end [Typhoon HIL — R5]	
Pre-charge circuit	Limit inrush: raise output to battery voltage before main contactor closes	DC out via resistor/DC/DC ramp	Pre-charge resistor + relay, or controlled voltage ramp of the DC/DC stage *(arch-dep.)*	State-machine controlled	Resistor burnout; timeout (battery not matching)	Pre-charge duration logging vs battery V; resistor thermal check	Essential
Discharge (bleed) circuit	De-energize output after stop; support welding detection	Output caps → discharge	Active dump or resistors	State-machine controlled	Bleeder open → output stays live	Measure output decay time after stop	Essential
IMD	Insulation resistance monitoring DC→PE; cable check & continuous monitoring	DC out + PE → status	Thresholds commonly expressed in Ω/V of operating voltage; e.g. <100 Ω/V → fault, >500 Ω/V → OK band *(per IEC 61851-23 design practice; verify edition)* [Gongyuan — R10]	Digital/analog status to MCU; result reported in CableCheck	Self-test failures; nuisance trips in humid conditions	Simulated insulation resistance calibrator; self-test button	Essential for DC
Ground/earth monitoring	Verify PE integrity before/during operation	PE circuit	PE continuity/ presence check	Digital input	Corroded earth stake in fixed installs	Loop resistance tests at commissioning	Essential
PLC modem / SECC	Vehicle digital link (SLAC + V2G protocol)	CP+PE carrier → Ethernet/IP inside charger	HomePlug Green PHY, QPSK/BPSK/ 256-QAM roaming per profile	Messages over internal bus to main controller	No SLAC match; noisy CP injection circuit; firmware mismatch to EV	PLC sniffer/log, SLAC logs (attenuation dB values), protocol traces	Essential for CCS
Control Pilot (CP) interface	PWM generation & CP voltage state detection; connector lock control	CP pin	±12 V levels; 1 kHz PWM; duty cycle signals available current in AC/ basic signalling	ADC + GPIO + PWM periph	Diode fault detection (EV side), dirty pins → wrong state	Pilot-state oscilloscope capture; state-transition tests	Essential
Proximity Pilot (PP)	Plug presence (and AC cable rating via resistor)	PP pin	Resistor-coded	ADC input	Broken resistor coding, dirty pins	State check on insertion	Arch-dep.
CCS2 connector & cable	Physical + signalling interface	—	DC pins plus CP/PP/PE; liquid-cooled cables for high current *(varies)*	—	Pin wear/ arcing, cable damage, latch failure	Visual + contact-resistance measurement; lock actuator test	Essential
Cooling system (fans/ pumps/ coolant)	Remove losses from modules & cable	—	Fan/air for lower power; water-glycol loops for high power	Pump/fan control + flow/ leak/temp sensors	Pump wear, coolant leaks, fan bearing failure, clogged filters	Flow switches, ΔT monitoring, leak detection loops	Arch-dep.
Energy meter	Billable kWh measurement	Output (or grid side) → kWh	Class-indexed meters per local metrology rules *(region-dep.)*	Modbus/UART/ internal	Calibration drift; tampering	Sealed calibration; meter self-diagnostics	Essential (billing)
HMI / display	User interface & status	User touch → controller	Touchscreen, LEDs	UART/CAN/ USB/internal			Arch-dep.

					Screen failure; sunlight washout	Frame-buffer tests; touch calibration	
RFID / NFC reader	User authentication	Card → ID	MIFARE/ISO 14443/15693 class	UART/SPI/ USB	Antenna damage; reader lockups	Test card set; read-range test	Optional
Ethernet / CAN / RS485 / cellular	Internal & external connectivity	—	Internal buses: CAN between boards; external: Ethernet, 4G/ LTE	Various	Signal integrity, modem drops	Bus error counters (CAN), ping/ throughput tests	Arch-dep.
Emergency stop & interlocks	Immediate safe shutdown; door/ access interlocks	Button → safety chain	Hard-wired safety circuit (independent of software best-practice)	Hardwired + logged	Contact corrosion; cable damage	Periodic E-stop function tests	Essential
Backend / OCPP link	Operator cloud: auth, pricing, telemetry, remote control	Charger → CSMS	OCPP 1.6/2.0.1 over Ethernet/ cellular *(feature-dep.)*	Main controller	Network outages; message queue overflow	OCPP message logs; offline-auth fallback testing	Arch-dep.

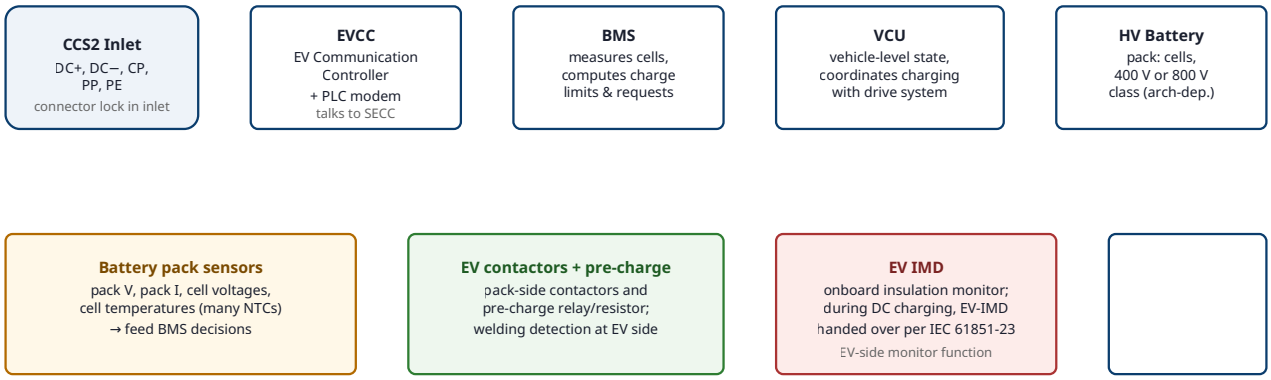
Table 3.1 — Charger-side component map. Ratings marked "illustrative" vary by product; verify with the specific datasheet before quoting.

3.1 Design variations to mention

- Single-gun vs dual-gun dispensers sharing one power cabinet (power sharing / dynamic allocation).
- Integrated (all-in-one) chargers vs split power-cabinet + dispenser architectures.
- Bidirectional (V2G-capable) units replace one-way DC/DC modules with bidirectional topologies and add grid-interaction logic.
- Solar + battery-buffered sites add a site controller and change the AC input path entirely.

4. EV-Side Hardware

Figure 12 — EV-side hardware relevant to DC fast charging



[TI SLUAB09 — R11] OBCfor AC charging;bypassed in DCfast charging BMS sets charging limits & current requests; EVCC carries them in V2G messages Who owns what (common split — details vary by OEM): • EV: inlet, connector lock (EV side), BMS limits/requests, EV contactors/pre-charge, EVCC+PLC modem, EV IMD, welding detection. • Charger: cable & plug lock, power conversion, EVSE-side IMD/cable check, pre-charge of output to battery voltage, session metering/billing. • Shared over protocol: EV notifies ready-states and requests (CurrentDemand); charger reports EVSE limits, current limits and status/fault codes. Note: the charger never "talks to the BMS directly" — it talks to the EVCC; the EVCC and BMS exchange information over the vehicle's internal network (e.g. CAN).

Figure 12. EV-side block diagram. Internal vehicle buses (CAN etc.) connect EVCC ↔ BMS ↔ VCU; only EVCC talks externally to the charger's SECC.

4.1 Component explanations

- **CCS2 charging inlet**: the vehicle-side socket with DC+, DC−, CP, PP and PE contacts; includes the EV-side connector lock actuator in many designs *(lock location — inlet vs cable — is architecture-dependent)*.
- **BMS**: measures cell voltages, pack current and temperatures; computes the safe operating window (max/min voltage, max current, max temperature) and decides the *current request* each second of the session. It is the source of truth for what the battery can accept.
- **VCU**: coordinates vehicle-level state (drive, charge, sleep), decides when charging is allowed, manages thermal preconditioning in some vehicles *(varies by OEM)*.
- **HV battery pack**: 400 V-class or 800 V-class cells (pack-level nominal voltage e.g. ~400 V or ~700–800 V, *architecture-dependent*); the reason chargers need a wide, programmable output voltage range.
- **Battery sensors**: pack voltage, pack current, per-cell voltages, many temperature points; feed the BMS's protection and request calculations.
- **EV contactors + pre-charge**: connect the pack to the inlet; EV pre-charges its inverter/link before closing main contactors; the EV performs its own welding detection. (Do not confuse EV-side pre-charge with EVSE-side pre-charge of the output to battery voltage — both exist, at opposite ends of the cable.)
- **EV insulation monitoring**: the vehicle has its own isolation monitor for driving; during DC charging per IEC 61851-23 practice, the EV-IMD is de-activated and the EVSE-side IMD takes over, with a defined handover sequence [TI SLUAB09 — R11] *(summary of standard practice — flag as such)*.

- **EVCC (EV Communication Controller)**: the vehicle's modem+protocol processor that runs SLAC and DIN 70121/ISO 15118 on the EV side. It is the *only* vehicle entity the charger communicates with over the cable; the BMS ↔ EVCC link is internal (e.g. CAN) *(typical)*.

5. EV ↔ Charger Communication: Complete Protocol Explanation

5.1 CCS2 connector pins

Figure 2 — CCS2 connector contact layout (vehicle inlet, front view, schematic)

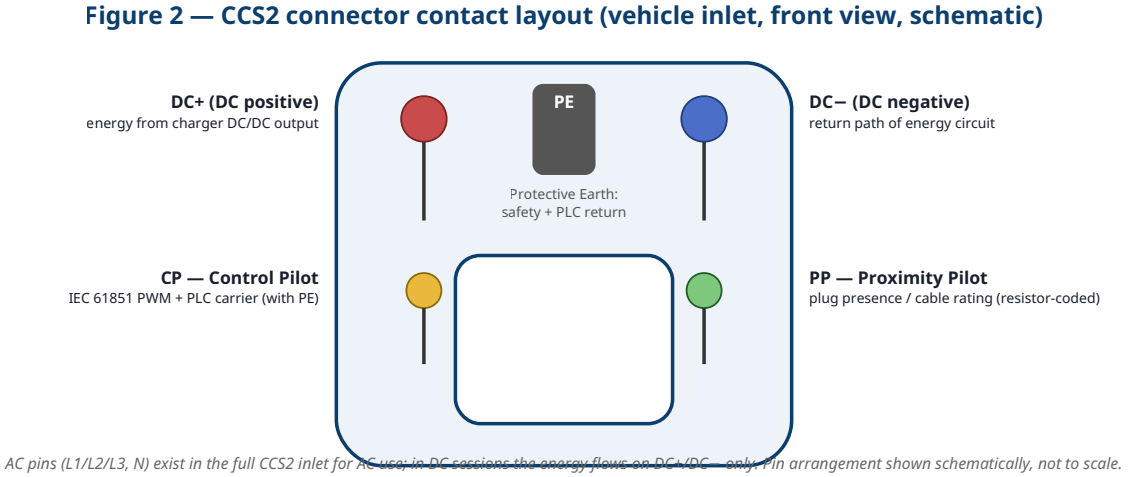


Figure 2. CCS2 contact functions. The CP line carries both the IEC 61851 PWM and the HomePlug Green PHY PLC signal (injected CP-to-PE).

5.2 Purpose of CP and PP

CP is the charger→vehicle "conversation wire": the EVSE applies a +12 V/−12 V square wave at 1 kHz (when appropriate), and the EV changes the *load resistance* it presents on that line to signal its state. The EVSE measures the resulting voltage level to detect the state. On top of this, in CCS DC charging, a PLC modem on both sides injects broadband data onto the CP–PE loop. **PP** tells the charger that a plug is physically inserted (and in AC charging, via a resistor code, the current rating of the cable). PP is a simple, low-speed signal; CP is the safety-critical signalling link [HDT CP/PP FAQ — R2] *(secondary summary)*.

5.3 Control Pilot states and PWM (IEC 61851-1)

Figure 3 — Control Pilot states (voltage levels) and 1 kHz PWM concept

CP states (nominal ±1 V)

[Geocadra summary — R1] State A: +12 V — no EV State B: +9 V — EV present (PWM) State C: +6 V — charging (PWM) State D: +3 V — ventilation (rare) State E: 0 V — EVSE error/no power State F: −12 V — pilot fault 1 kHz PWM on CP (EVSE-generated) +12 V−12 V duty cycle D (e.g. 40%) 1 ms period (1 kHz) In States B/C the EVSE transmits this square wave. In CCS the duty cycle's current-advertising role is largely superseded by the digital session

(EV/charger exchange limits precisely via ChargeParameterDiscovery / CurrentDemand). How the EV changes state: The EV switches resistances (e.g. 2.74 kΩ → B, 882 Ω → C in the IEC 61851-1 scheme) between CP and PE, plus a diode so negative half-waves are blocked — the EVSE also checks this diode to detect a non-genuine EV/pilot fault. Secondary-summary levels; verify exact resistor values against IEC 61851-1 when designing (values above are the widely published nominal ones).

Figure 3. CP concept: EVSE-generated ±12 V, 1 kHz PWM; EV loads the line with resistors to announce State B/C/D; EVSE reads voltage; diode check detects pilot-line faults.

State table

State	Nominal CP voltage	Meaning	EVSE action
A	+12 V (no PWM)	No vehicle connected	Idle
B	+9 V (PWM present)	Vehicle detected, not ready	Connector may lock; no power yet
C	+6 V (PWM present)	Vehicle ready / charging (closes its switch)	Energize power contacts
D	+3 V (PWM present)	Ventilation required (rare; legacy lead-acid scenario)	Energize with ventilation requirement
E	0 V	EVSE error / no power	Stop power
F	−12 V	Pilot fault (e.g. diode short)	Stop power, latch fault

Source: [Geocadra IEC 61851 summary — R1], [HDT — R2] (secondary summaries of IEC 61851-1; confirm against the standard itself for design work).

PWM duty cycle: in basic IEC 61851 (AC) charging the PWM duty cycle advertises the maximum current available (e.g. specific % bands map to ampere ranges). In **CCS DC** charging the duty cycle still exists as part of CP behaviour, but the actual current limits are negotiated digitally in the PLC session (ChargeParameterDiscovery / CurrentDemand messages) — a distinction interviewers like to probe [Lewandowski PLC-over-IEC-61851 paper — R9; Geocadra — R1]. ಸರಳವಾಗಿ: CP ತಂತಿ "ನಾನು ಎಷ್ಟು ಕರೆಂಟ್ ಕೊಡಲು ಸಾಧ್ಯ" ಎಂಬುದನ್ನು PWM ಮೂಲಕ ಹೇಳುತ್ತದೆ; ನಿಖರವಾದ ಮೌಲ್ಯಗಳನ್ನು PLC ಡಿಜಿಟಲ್ ಸಂಭಾಷಣೆಯಲ್ಲಿ ಚರ್ಚಿಸಲಾಗುತ್ತದೆ.

5.4 Power Line Communication (PLC) and HomePlug Green PHY

ISO 15118-3 (and its DIN 70121 predecessor in practice) specifies **HomePlug Green PHY (HPGP)** as the physical/data-link layer: an OFDM-based power-line modem technology operating in the ≈2–30 MHz band, injected between the CP line and PE on both the EV (EVCC) and charger (SECC) sides. HPGP is a low-power variant of HomePlug AV, designed for low-energy, high-reliability links [ISO 15118-3 summary — R4; EV Range PLC explainer — R8].

5.5 SLAC — Signal Level Attenuation Characterisation

Because the CP/PE medium is shared, before any digital session the EVCC and SECC must discover each other and pair. **SLAC** is that matching procedure: the EV sends (as CM_SLC_PARAM) discovery frames; the EVSE replies with attenuation (sound) profiles; the EV picks the matching EVSE, they exchange matching confirmation (CM_SLAC_MATCH) and then establish the logical network with keys. After SLAC comes link establishment (CM_SETKEY etc.), IP setup (SLAAC), TCP, TLS (if used) and then the V2G application protocol. SLAC verifies the connection path so that the right car talks to the right charger [ISO 15118-3 summary — R4].

5.6 EVCC, SECC and the protocol stack

Figure 4 — Protocol stack across EV and charger (CCS2 DC, ISO 15118-2 flavour)

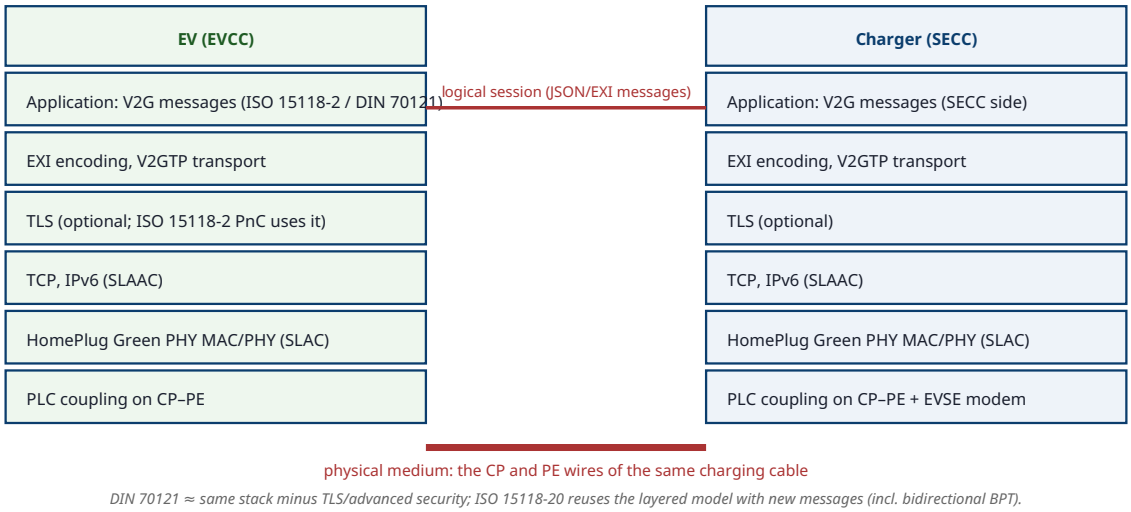


Figure 4. Layered view: SLAC/HPGP at the bottom, TCP/IP in the middle, EXI-encoded V2G messages at the top.

- **EVCC** = EV Communication Controller (vehicle side). **SECC** = Supply Equipment Communication Controller (charger side). They are the two endpoints of the PLC session.
- **DIN 70121**: the earlier, DC-only CCS protocol. Supports the core charging messages but lacks full PKI-based Plug & Charge and end-to-end TLS security; still the most widely deployed baseline for DC CCS [Jointcharging DIN 70121 explainer — R6]. *One widely republished summary claims DIN 70121 "supports simple TLS" — the majority description (R4, R6, renco R-31) says TLS/Plug&Charge security is **not** part of DIN 70121; treat exact security scope as version-dependent and verify against the DIN document if it matters.*
- **ISO 15118-2**: adds Plug & Charge (PKI certificates), TLS transport, AC and DC charging, scheduling [ISO 15118-2 OBP — R5].
- **ISO 15118-20**: the 2022 second generation — new message set, bidirectional power transfer (BPT/V2H etc.), improved plug & charge, AC/DC/WPT (wireless) modes [ISO 15118-20 doc — R14]. Deployment is ramping in 2024–2026 vehicles.
- **Not every vehicle or charger runs every protocol**: vehicles typically advertise DIN 70121 and/or ISO 15118-2 (rarely -20 yet); the charger picks the mutually supported version in `SupportedAppProtocol`. Older EVs may only do IEC 61851 CP (no PLC) on AC, and some non-CCS DC standards (e.g. GB/T, NACS variants) exist outside the CCS2 world.

5.7 High-level messages, limits and negotiation (what "the protocol" actually says)

The V2G application layer is a request/response message exchange. The DC-charging message flow below is from the Typhoon HIL reference implementation documentation of ISO 15118-2 [R5]:

Message (Req/Res pair)	Phase	What it does, in plain English
`supportedAppProtocol`	Setup	EV and charger agree which protocol version to speak (e.g. DIN 70121 vs ISO 15118-2).
`SessionSetup`	Setup	Charger assigns a unique SessionID for this session.
`ServiceDiscovery`	Identification	EV asks what the charger offers (energy modes, payment options).
`PaymentServiceSelection`	Identification	EV picks the authentication method (e.g. Plug & Charge vs external card).
`Authorization` (+ PaymentDetails / Certificate… for PnC)	Identification	Authenticate the user/vehicle; anti-replay checks.
`ChargeParameterDiscovery`	Target setting	Exchange of max/min voltage and current limits of both sides + energy schedules.
`CableCheck`	DC only	EV tells charger it is ready/locked; charger runs insulation test of the DC cable/output and reports status.

`PreCharge`	DC only	Charger raises its output voltage to the EV's battery voltage (EV sends target voltage + max inrush current) so contactor closure sees ~0 V difference.
`PowerDelivery (Start)`	Energy transfer	The "go" switch: EVSE energizes output; EV starts drawing current.
`CurrentDemand` (loop, ~every second or so)	Charging loop	EV continuously requests target voltage/current and receives EVSE-present limits, status and fault info. This is where dynamic adjustment happens.
`PowerDelivery (Stop)`	End	EV signals pause/stop; charger ramps down.
`WeldingDetection`	End	Contactor welding detection mechanism per IEC 61851.
`SessionStop`	End	Closes the communication session cleanly.

Charging limits vs actual output: each side owns limits — the EV declares what its battery can accept (EVMaximumVoltageLimit, EVMaximumCurrentLimit…), the charger declares what it can deliver (EVSEMaximumVoltageLimit, EVSEMaximumCurrentLimit…). The *actual* operating point is the EV's current request inside the intersection of those limits, which the charger's control loop then realizes on the output. Status (EVSEStatusCode, e.g. `EVSE_IsolationMonitoringActive`, error codes) travels in the responses [Typhoon HIL — R5; ISO 15118-2 — R5b].

Timeouts and faults: every message has a performance time; if the peer (or the PLC link) doesn't respond in time, the sequence fails and the session is terminated safely. Loss of communication causes the EVSE to stop power delivery (e.g. after the defined sequence/timeout in IEC 61851-23) — the exact timeout ladder is standard-defined and version/product dependent.

Security / Plug & Charge: ISO 15118-2 with TLS + PKI lets the vehicle authenticate with a contract certificate — no card or app; billing is bound to the vehicle identity. ISO 15118-20 extends and modernizes this. DIN 70121 sessions typically use external authentication (RFID/app) instead [R6, R8].

5.8 Full communication sequence diagram

gure 5 — EV ↔ Charger communication sequence (CCS2 DC, ISO 15118-2-style; DIN 70121 = same minus TLS/Pn

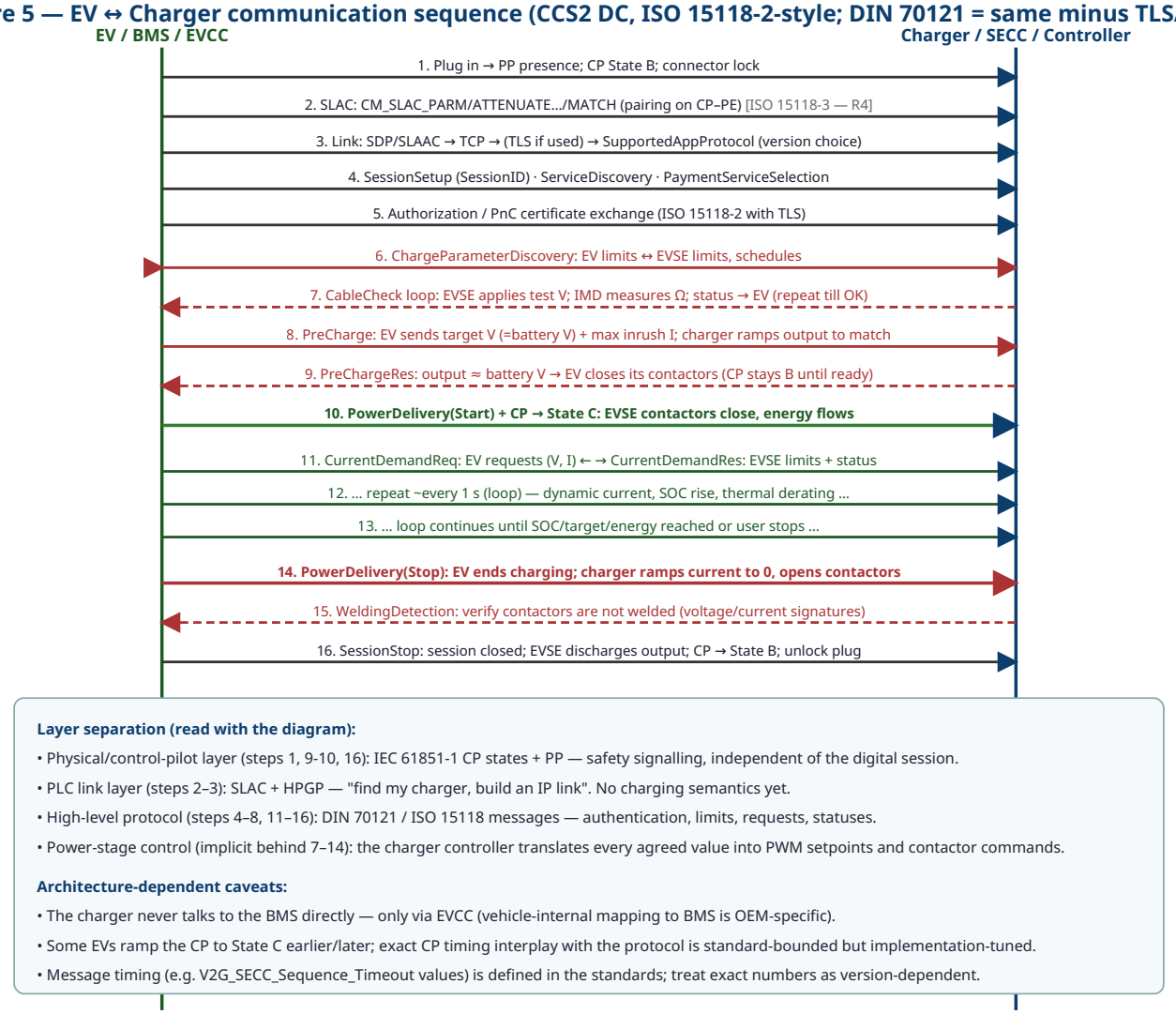


Figure 5. Numbered EV↔charger sequence. Green = energy-transfer loop; red = safety-critical steps. Sequence per ISO 15118-2 documentation [R5].

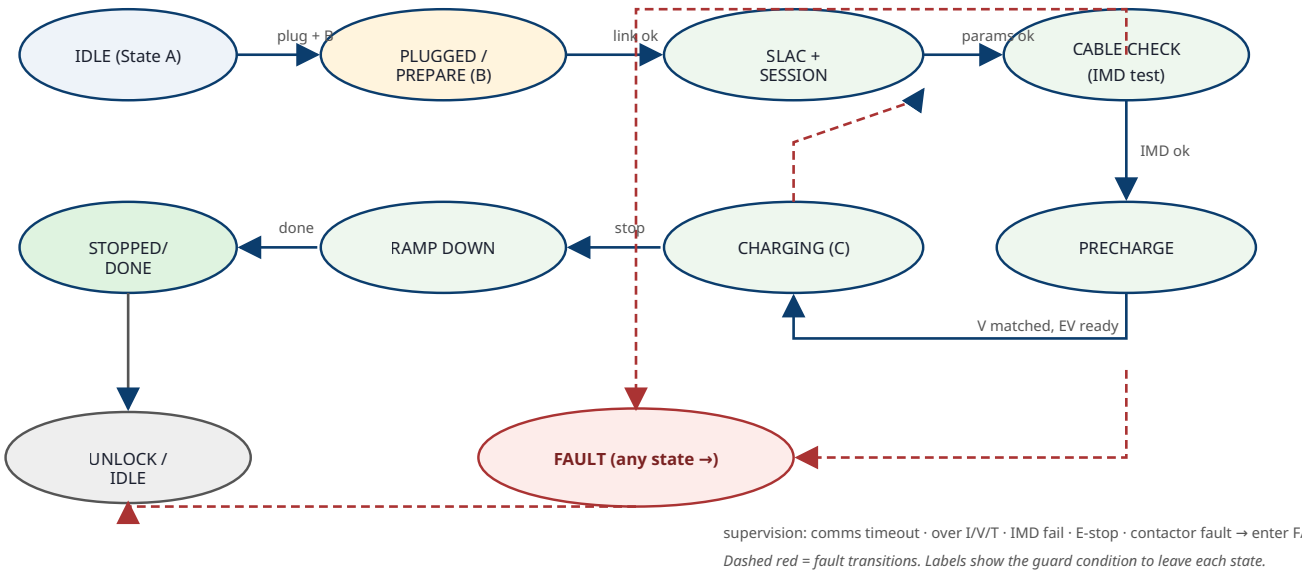
6. Charging Sequence: Step-by-Step

The 17 stages you asked about, mapped to what happens electrically and in software. (Stage numbers below match your list.)

#	Stage	What happens (technical)	Simple language
1	Cable connection	User plugs CCS2 gun; PP contact made; charger detects plug presence; vehicle wakes the EVCC (CP line power / PLC wake).	"ಪ್ಲಗ್ ಸಿಕ್ಕಿಸಿದೆ" — the plug goes in.
2	Connector & safety checks	Connector lock actuation, plug temperature check, EVSE self-tests, PE presence verified.	The charger checks the plug is seated, locked and healthy.
3	CP state detection	EVSE sees State B (+9 V PWM); diode check passed; PWM transmitted per IEC 61851-1 [R1, R2].	The car "raises its hand" on the CP wire.
4	PLC link establishment	EVCC modems (both sides) detect the link; CM_SLAC_PARM.REQ etc. begin.	The two computers find each other through the cable.
5	SLAC matching	Attenuation characterisation, CM_SLAC_MATCH, key exchange; link up; IPv6/TCP established.	They agree "I am talking to the right charger/car".
6	Protocol/session negotiation	SupportedAppProtocol → version; SessionSetup → SessionID; ServiceDiscovery; PaymentServiceSelection; Authorization (or PnC).	Handshake: who are you, how do we pay.
7	Capability exchange	ChargeParameterDiscovery: EV max/min V, max I, energy needs vs EVSE output window (e.g. 150–1000 V, 0–500 A *illustrative*).	Both sides state their limits honestly.
8	Parameter negotiation	Target SOC, energy request, schedule (SAScheduleTuple) agreed; EVSE declares it can support the plan.	Deal: this much energy by this time.
9	Insulation & safety checks	CableCheck loop: EVSE applies test voltage through IMD; insulation resistance measured against thresholds (Ω/V rule of thumb, edition-dependent); EV confirms lock/ready [R10].	Leakage test of the cable before real power.
10	Pre-charge / voltage matching	PreCharge: charger regulates output to EV battery voltage within requested inrush limit; EVSE output ≈ pack voltage.	Match the two voltages so closing the switch doesn't spark.
11	Contactor operation	EV closes its contactors (ΔV≈0); EVSE closes output contactors; CP → State C at the defined point.	The switches close.
12	Controlled power delivery	PowerDelivery(Start) confirmed; charger current ramp per request; control loops track (V, I) setpoints.	Power flows, under control.
13	Continuous monitoring	Every loop: output V/I, temperatures, IMD, coolant, DC-link plausibility; watchdogs; anomaly counters.	Constant health checks.
14	Dynamic adjustment	Each CurrentDemand message updates the request; EV may reduce current (thermal, cell-voltage, balancing reasons); charger follows within milliseconds (electrical loop) + seconds (message loop).	The car keeps steering; the charger follows.
15	Stop request	EV sends PowerDelivery(Stop) (target reached, user stop, or fault), or EVSE terminates on fault/backend stop.	Someone says "enough".
16	Ramp-down & safe disconnect	Current → 0, contactors open, welding detection, output discharge, CP → State B, unlock, SessionStop.	Wind down, unlock, unplug.
17	Fault handling / termination	On any fault: per-severity response (derate → pause → emergency stop); session terminated, reason logged (EVSEStatusCode / logs), backend notified.	If something is wrong, stop safely and record why.

6.1 State-machine diagram

Figure 6 — Simplified charger session state machine (educational; real firmware adds many more guards)



Educational simplification. Production state machines add sub-states (e.g. pre-charge retry, ventilation handling, EVSE re-negotiation) and per-standard timeout ladders.

Figure 6. Charging session state machine (simplified educational model).

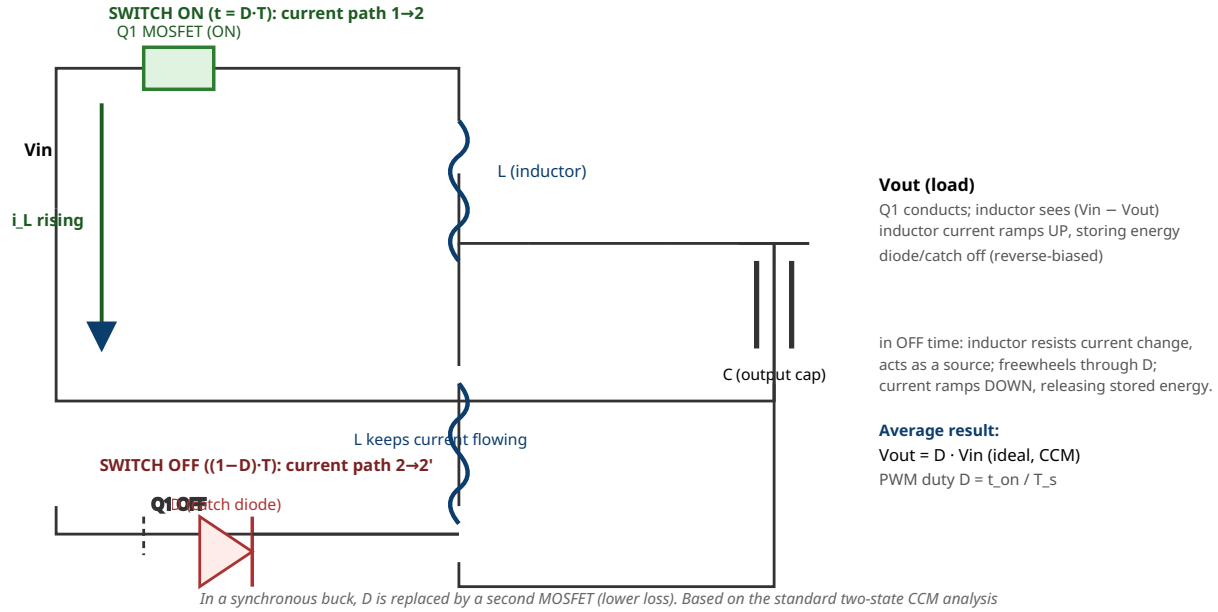
7. Buck Converter: Full Technical Explanation

7.1 What it is and why it is used

A **buck (step-down) converter** is a switching DC/DC converter that produces an output voltage *lower* than its input, with high efficiency (typically 85–97% *typical, depends on design*) by switching an inductor at high frequency instead of burning voltage in a linear regulator. ಬಕ್ ಕನ್ವರ್ಟರ್ ಹೆಚ್ಚಿನ ವೋಲ್ಟೇಜನ್ನು ಕಡಿಮೆ ವೋಲ್ಟೇಜಾಗಿ ಪರಿವರ್ತಿಸುತ್ತದೆ — ಶಕ್ತಿ ವ್ಯರ್ಥವಾಗದಂತೆ ಇಂಡಕ್ಟರ್ ಮತ್ತು ಸ್ವಿಚ್ ಬಳಸಿ. It is used everywhere a lower rail is needed from a higher one: 12 V→1.2 V CPU cores, 24 V→12 V auxiliaries, and inside charger systems (auxiliary rails, and the underlying principle of controlling average output voltage by PWM in many power stages).

7.2 Circuit and current paths

Figure 7 — Buck converter: circuit and ON/OFF current paths



[TI SLVA057 — R3].

Figure 7. Buck converter ON and OFF current paths with labelled components: Q1 (switch), D (catch/freewheel diode), L, C, load.

7.3 Operation cycle and component behaviour

- **ON phase** (Q1 closed): the inductor sees ($V_{in} - V_{out}$); current ramps up linearly ($dI/dt = (V_{in} - V_{out})/L$); energy is stored in the magnetic field; the capacitor supplies/absorbs ripple current so the load voltage stays nearly constant.
- **OFF phase** (Q1 open): the inductor *opposes the change*, its voltage flips polarity to keep current flowing, and the catch diode (or synchronous MOSFET) carries the freewheeling current; the inductor releases energy into the load while current ramps down.
- **Steady state (CCM)**: the inductor's volt-second balance — $V_{in} \cdot t_{on} = V_{out} \cdot t_{off}$ — gives the ideal conversion ratio $\mathbf{V_{out} = D \cdot V_{in}}$, i.e. $\mathbf{D = V_{out}/V_{in}}$ [TI SLVA057 — R3].

7.4 Worked example (with stated assumptions)

> Assumptions: ideal components (no V_{DS} drop, no diode drop), continuous conduction mode, steady state.
> $V_{in} = 400$ V, target $V_{out} = 200$ V, $f_{sw} = 100$ kHz, $L = 200$ µH, $R_{load} = 4\ \Omega$.
>
> 1) Duty cycle: $D = V_{out}/V_{in} = 200/400 = 0.5$

2) Output current: $I_o = V_{out}/R = 200/4 = 50$ A; output power $P_o = 200 \times 50 = 10$ kW

3) On-time: $t_{on} = D/f_{sw} = 0.5 \times 10$ µs $= 5$ µs

4) Inductor ripple: $\Delta I_L = (V_{in} - V_{out})\cdot D\ /(L\cdot f_{sw}) = (200\text{ V} \times 0.5)/(200\text{ µH} \times 100\text{ kHz}) = 100/20 =$ 5 A peak-to-peak

5) Peak inductor current: $I_{L,pk} = I_o + \Delta I_L/2 = 50 + 2.5 = 52.5$ A

Output voltage ripple (illustrative, ideal CCM): for a pure capacitance model, $\Delta V_{out} \approx \Delta I_L\ /(8\cdot f_{sw}\cdot C_{out})$. With ΔI_L = 5 A, f_{sw} = 100 kHz, C_{out} = 2 mF → $\Delta V_{out} \approx 5/(8\times10^5\times2\times10^{-3}) \approx$ **3.1 mV** (real designs add ESR effects; this is the textbook ideal term).

7.5 Feedback control and the embedded role

A controller measures V_{out} (and usually I_L), compares it with the reference, and a compensator (PI/PID or type-II/III) adjusts the PWM duty D. In an MCU-based design: ADC samples the feedback → control law computed → compare/timer registers updated → PWM peripheral outputs → gate driver amplifies/isolates → power switch. The gate driver is essential because MCU pins (≈3.3 V, mA) cannot charge the large gate capacitance of power MOSFETs (needs amps at 10–15 V_{GS}, isolated when the switch sits on a high-voltage node) [TI SLUA618 — R12].

7.6 Losses, efficiency, thermal and protection

- **Conduction loss**: I^2R — $R_{DS(on)}$ of the MOSFET, DCR of the inductor, ESR of capacitors.
- **Switching loss**: voltage-current overlap during each transition, $\approx \tfrac{1}{2}\cdot V\cdot I\cdot(t_r+t_f)\cdot f_{sw}$; rises with frequency (why SiC helps: faster, lower-overlap switching at HV).
- **Other**: gate-charge loss, diode reverse recovery (non-synchronous), core (hysteresis + eddy) loss in L.
- **Thermal**: losses → junction temperature; heat-spreading and cooling design required; thermistors on hot spots.
- **Protection**: over-current (cycle-by-cycle current limit, desat/hiccup), over/under-voltage, thermal shutdown, soft-start to limit inrush.

7.7 Where buck fits in EV charging

Real DC fast chargers usually regulate the *battery-side* output with an isolated stage (LLC, DAB, phase-shifted full bridge) rather than a literal non-isolated buck. But: (a) the *control principle* (PWM duty + inductor + feedback loop) is identical, and interviewers use buck/boost as the litmus test; (b) the charger's DC-link is created by a **PFC boost** stage (§8); (c) auxiliary power supplies inside the charger are small bucks; (d) some non-isolated or modular topologies (e.g. buck-derived stages, active-clamp buck) appear in specific designs. *(Architecture-dependent.)*

8. Boost Converter: Full Technical Explanation

8.1 What it is

A **boost (step-up) converter** produces an output voltage *higher* than its input, by using the inductor's "kick-back": when the switch opens, the inductor voltage flips and *adds* to the input voltage, charging the output capacitor above V_{in}. ಬೂಸ್ಟ್ ಕನ್ವರ್ಟರ್ ಕಡಿಮೆ ವೋಲ್ಟೇಜನ್ನು ಹೆಚ್ಚಿನ ವೋಲ್ಟೇಜಾಗಿ ಪರಿವರ್ತಿಸುತ್ತದೆ — ಇಂಡಕ್ಟರ್ ಸಂಗ್ರಹಿಸಿದ ಶಕ್ತಿಯನ್ನು ಬಿಡುಗಡೆ ಮಾಡಿ.

8.2 Circuit and current paths

Figure 8 — Boost converter: circuit and ON/OFF current paths

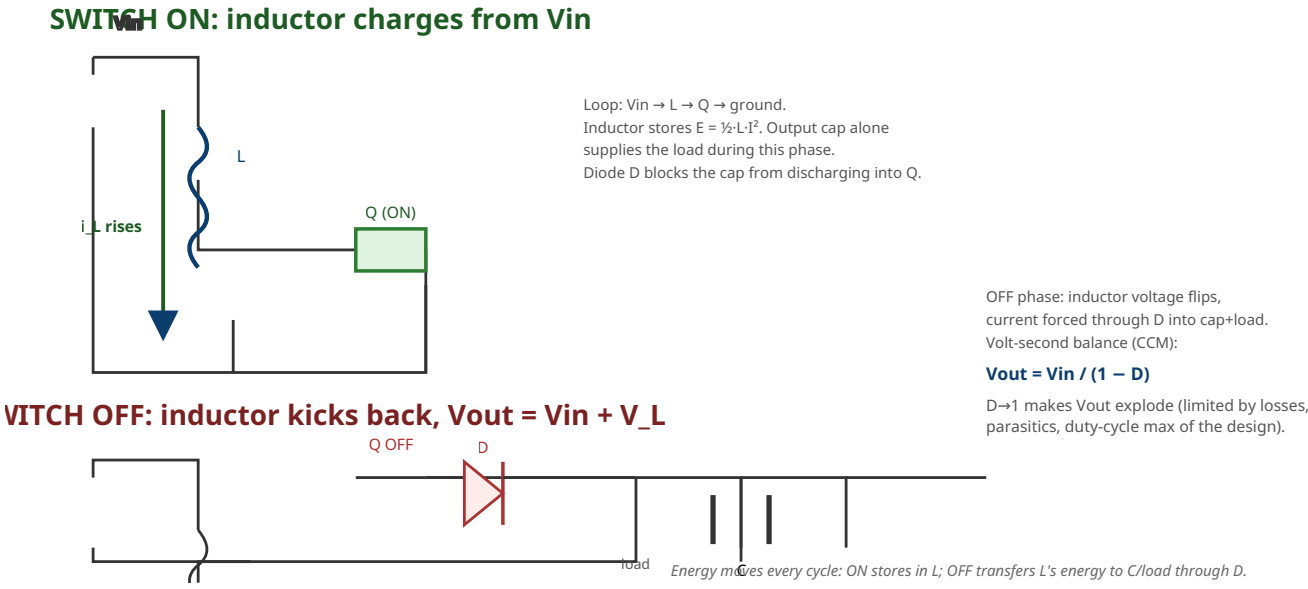


Figure 8. Boost converter ON/OFF paths: inductor charging, then inductor + input series into the output through the diode.

8.3 Operation

- **ON**: current builds in L from V_{in}; load is fed by C; D blocks back-discharge.
- **OFF**: L's current cannot stop instantly; L's voltage reverses to keep current flowing, so node after L jumps to V_{out}; current flows V_{in} → L → D → C/load. Energy per cycle from L is dumped into the output.
- **Volt-second balance** (inductor average voltage = 0 in steady state): $V_{in}\cdot D\cdot T = (V_{out} - V_{in})\cdot(1-D)\cdot T \rightarrow \mathbf{V_{out} = V_{in}/(1-D)}$ (ideal CCM) [ST AN6378 PFC-boost context — R15; standard theory].

8.4 Worked example (with assumptions)

> Assumptions: ideal, CCM.
> $V_{in} = 230$ V AC rectified → treat DC bus target 400 V from a 230 V DC input (grid-side PFC scenario).
> 1) $D = 1 - V_{in}/V_{out} = 1 - 230/400 = 0.425$
> 2) If $P_{out} = 4$ kW at $V_{out} = 400$ V → $I_o = 10$ A; input (inductor) average current $I_L = P/(V_{in}\times\eta)$, $\eta\approx1$ ideal → $4000/230 \approx 17.4$ A
> 3) With $f_{sw} = 50$ kHz, $L = 500$ µH: $\Delta I_L = V_{in}\cdot D/(L\cdot f_{sw}) = 230\times0.425/(500\ \mu\times50\ \text{k}) = 97.75/25 \approx 3.9$ A p-p
> 4) Peak switch current $= I_L + \Delta I_L/2 \approx 17.4 + 1.95 \approx 19.4$ A (sizes the MOSFET and D).

8.5 Feedback, sensing, limitations, protection

- **Feedback**: measure V_{out} (and often inductor/input current for current-mode control); adjust duty D. Note boost is non-minimum-phase (duty ↑ immediately reduces the average current into the output before V_{out} rises) — control design is subtler than buck.
- **Limitations**: practical duty ceilings (~0.7–0.9 for simple boosts) due to losses; diode reverse-recovery; high peak currents; poor transient handling at light load (DCM); right-half-plane-zero dynamics.
- **Protection**: switch over-current, output over-voltage (the cap can rise quickly if load disconnects — OVP is critical), inrush/soft-start, thermal.
- **Relevance to EV chargers**: the PFC front end of nearly every DC fast charger is a **boost-type rectifier** (single-phase boost PFC, or 3-phase variants such as the Vienna rectifier — a boost-derived topology) that shapes grid current and creates the ~800 V DC link *(topology-dependent)*. Boost concepts also appear in bidirectional stages (battery→grid direction), in on-board charger PFCs, and in auxiliary supplies.

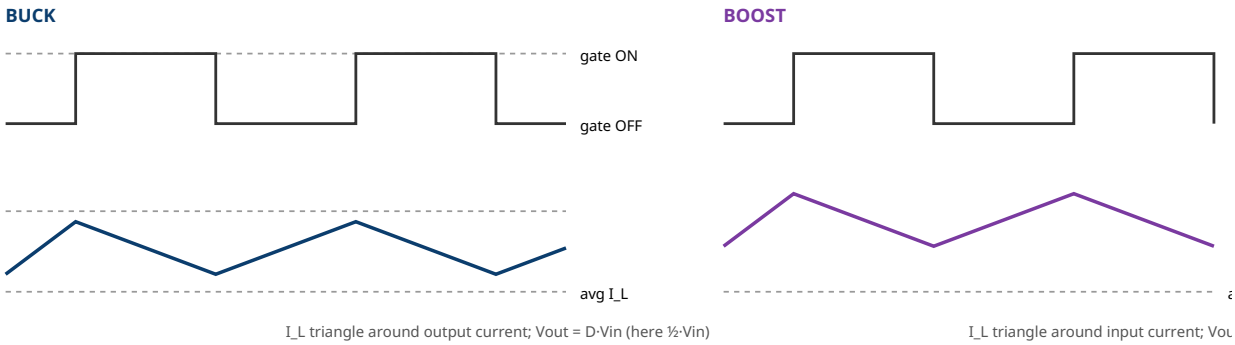
9. Buck vs Boost vs Buck-Boost

9.1 Comparison table

Criterion	Buck	Boost	Buck-Boost (inverting or 4-switch)
Voltage behaviour	$V_{out} < V_{in}$ (step-down)	$V_{out} > V_{in}$ (step-up)	V_{out} above, below (4-switch) or inverted magnitude of V_{in}
Core components	Series switch, catch diode / sync FET, inductor, output cap	Shunt switch at input side, series diode to output, inductor, output cap	Two switches + diodes (4-switch version), inductor; inverted polarity (classic) or same polarity (4-switch)
Duty relation (ideal CCM)	$V_{out} = D \cdot V_{in}$	$V_{out} = V_{in}/(1-D)$	$\lvert V_{out}\rvert = V_{in} \cdot D/(1-D)$
Efficiency notes	Direct energy path in ON phase; among the best simple topologies; sync rectification removes diode loss	All output energy passes through the diode/switch path; higher peak currents; reverse recovery matters	Two conversions per cycle (buck+boost cascade in 4-switch version) → more loss stages; extra switches
Typical use cases	Point-of-load rails, 48→12 V, aux supplies, battery discharging to lower bus	PFC pre-regulators, battery→HV bus raise, LED drivers, PV micro-inverters	Battery-powered devices needing a rail that can sit above or below the battery; USB-PD power roles
Advantages	Simple, efficient, cheap, excellent transient response	Simple, can generate HV from LV; natural PFC stage	Full range flexibility around V_{in}
Limitations	Cannot exceed V_{in}; input current pulsed (needs input filter)	Cannot go below V_{in}−(diode); OVP risk if load disconnects; control trickier (RHP zero)	More parts, lower efficiency, both switches see high current

9.2 Buck vs boost waveforms

Figure 9 — Buck vs boost: gate signal and inductor current in CCM (same duty D = 0.5, ideal)



Key waveform differences: in buck, the switch is in **series** with the input → input current is chopped, output sees smooth L-C filtered current. In boost, the inductor sits **in front** → input current is naturally continuous (great for PFC), but the switch current equals the inductor current and the diode current is pulsed into the output. Output ripple is larger; the output capacitor carries the load during the ON phase.

Sketches are idealized CCM shapes for concept illustration, not simulation output.

Figure 9. Comparison of switching behaviour and inductor-current shapes.

9.3 Real DC fast chargers: beyond simple buck/boost

- **Isolation**: DC charging practice requires galvanic isolation between grid and vehicle — a plain buck/boost has no isolation, so real chargers use transformers (LLC, DAB, PSFB) or modular isolated stages. *Standard practice; the isolation requirement details live in IEC 61851-23 — verify edition.*
- **Bidirectionality**: V2G-capable chargers need power flow in both directions → topologies that reverse flow (DAB, bidirectional LLC/boost-derived) and control software for discharge sessions (ISO 15118-20 BPT).

- **Modularity/interleaving**: parallel modules share current (interleaved phases cut ripple), letting one cabinet scale from 50 kW to 500 kW+; power blocks can be switched on/off for efficiency at part load.
- **Why topology depends on requirements**: wide output voltage range (150–1000 V) needs topologies with wide gain or module reconfiguration (series/parallel stacking); isolation and safety rules; efficiency targets (SiC vs IGBT); EMC; cooling; and cost. A single "buck or boost" circuit cannot meet all of these — that's the mature answer to give in an interview.

10. Control System and Embedded Firmware

10.1 Closed-loop control view

Figure 10 — MCU/ADC/PWM/sensor feedback-control loop in a charger

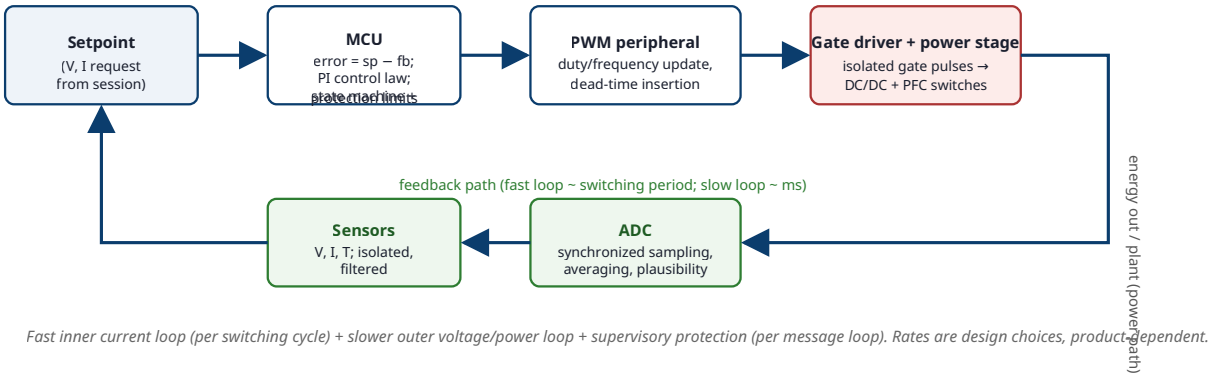


Figure 10. Feedback-control structure from setpoint to power stage and back through sensing/ADC.

10.2 Firmware building blocks

- **MCU architecture & peripherals**: 32-bit MCU families with fast ADCs (with simultaneous multi-channel sampling), high-resolution PWM timers (with dead-time), CAN/UART/SPI/I2C for sensors and boards, Ethernet for backend/PLC boards. Multiple MCUs often split: one for safety/power loops, one for comms/session, one for HMI.
- **ADC sampling**: synchronized to PWM (sample at counter underflow/zero-vec) to avoid switching noise; oversampling and median filters; per-channel plausibility checks (range, rate-of-change, cross-sensor consistency).
- **PWM generation**: duty from control law; dead-time insertion so high/low switches never conduct together; phase-shift for interleaved/bridge legs; min/max duty clamps.
- **Gate-driver control**: enable/interlock lines, desat fault handling (immediate hardware trip + logged latch-up), brown-out behaviour (drivers must default off).
- **Feedback loops**: inner current loop, outer voltage loop, power-limit loop (grid connection limits), thermal derating loop (module temperature → current ceiling).
- **State machine**: session lifecycle states as in Figure 6, with timeout guards on every transition.
- **Communication stack**: CP/PWM handling, SECC/PLC session handling (often on a dedicated SoC), OCPP client, internal CAN between control boards.
- **Fault monitoring & protection**: hardware comparators for cycle-level trips + software supervision with severity classes (log-and-derate vs safe-stop vs latch-out requiring service).
- **Watchdog & diagnostics**: independent watchdog + window watchdog; self-tests (RAM/flash/CRC of code, sensor tests) at boot and periodically; persistent event log in non-volatile memory with timestamps.
- **RTOS vs bare-metal**: bare-metal superloop with interrupt-driven control possible for simple units; RTOS (or at minimum prioritized ISRs + cooperative tasks) when many independent timers/stacks coexist. The safety-critical loops should never depend on non-real-time tasks (display, networking).
- **Event logging & debugging**: structured event/error codes, black-box session traces (V/I envelope, faults), SWD/JTAG in development, and remote log retrieval via backend.

- **Calibration & testing**: sensor offset/gain calibration stored per unit; HIL (§11.5) and bench tests with programmable loads/sources; type tests per regional regulation.

10.3 Conceptual firmware flowchart + educational pseudocode

```
/* EDUCATIONAL PSEUDOCODE — illustrates structure, NOT production-ready safety firmware.
   A real charger controller requires certified functional-safety design, redundancy,
   hardware interlocks and conformance testing beyond this sketch. */

const uint32_t LOOP_1MS = 1000u;   /* fast control tick */
static SessionState g_state = ST_IDLE;

void main(void)
{
    hw_init();          /* clocks, GPIO, ADC, PWM (outputs disabled), CAN, watchdog */
    self_test();        /* RAM/flash CRC, sensor sanity; fail → ST_FAULT_SERVICE */
    rtos_or_scheduler_start();
}

/* ---- 1 ms control ISR: sensing + loops + protection (always highest priority) ---- */
void ISR_control_1ms(void)
{
    SensorSet s = adc_read_all();            /* V_out, I_out, T_modules, V_dclink ... */
    if (!plausible(s)) { fault(RAW_SENSOR_IMPLAUSIBLE); }

    if (hw_trip_active()) { enter_safe_state(); }        /* gate-driver desat, E-stop, HW OCP */

    switch (g_state)
    {
      case ST_IDLE:      keep_output_off(); break;

      case ST_CABLECHK:  if (!imd_test_running()) start_imd_test();
                         if (imd_result_ok())       g_state = ST_PRECHARGE;
                         else if (timeout())         fault(INSULATION_FAIL);
                         break;

      case ST_PRECHARGE: ramp_output_voltage(ev_request.voltage, ev_request.i_limit);
                         if (abs(v_out - ev_request.voltage) < MATCH_EPS && settled())
                             g_state = ST_ENERGIZED;           /* ready for contactor close */
                         else if (timeout())          fault(PRECHARGE_TIMEOUT);
                         break;

      case ST_CHARGING:  v_ref = clamp(ev_request.voltage, 0, MY_VMAX);
                         i_ref = clamp(ev_request.current,  0, min(MY_IMAX, thermal_limit()));
                         pi_loop_run(&i_loop, i_ref, s.i_out);   /* inner loop */
                         pi_loop_run(&v_loop, v_ref, s.v_out);   /* outer loop */
                         break;

      case ST_RAMPDOWN:  ramp_current_to_zero();
                         if (i_out < ZERO_EPS) { open_contactors(); g_state = ST_WELDCHK; }
                         break;

      case ST_FAULT:     enter_safe_state();        /* PWM off, contactors open, bleed on */
                         break;
      default: break;
    }
    comm_timeout_check();   /* CP state, PLC session, internal CAN heartbeats */
    watchdog_kick();
}

/* ---- message-task: V2G session (runs on SECC-side; requests forwarded to 1 ms task) ---- */
void on_current_demand(const CurrentDemandReq *r)
{
    if (state_ok_to_accept()) { ev_request = *r; }        /* clamped later in control ISR */
    else { respond(EVSE_NOT_READY); }
}
```

```
void on_power_delivery_stop(void)   { g_state = ST_RAMPDOWN; }
void on_comm_timeout(void)          { fault(COMM_LOSS); }   /* → safe state per standard */
```

> **Say this in the interview:** the separation of a hard-real-time 1 ms control/protection task from a softer message/session task is the core embedded-architecture insight of a charger controller. Also emphasise that CP state handling has *hardware-level* responses (comparators, contactor driver enable lines) that work even if software hangs.

11. MATLAB/Simulink and Testing

11.1 Modelling a buck converter

MathWorks' own Simscape Electrical "Buck Converter" example models a switching supply converting a 30 V DC input into a regulated 15 V output; you use the model to size L and C and to design the feedback controller, choosing continuous or discrete controllers to study discretisation effects; modelling the switches as MOSFETs (not ideal switches) captures on-resistance and gate-timing effects [MathWorks Buck Converter example — R16]. Steps for a beginner model:

1. Open Simscape → Electrical → build: DC Voltage Source → MOSFET (or Controlled PWM + Half-Bridge Driver blocks, as the example uses) → series inductor → parallel capacitor + resistive load.
2. Add the catch diode (nonsynchronous) or second MOSFET (synchronous).
3. PWM: use a PWM Generator or compare a sawtooth carrier with your duty signal.
4. Measure: Voltage Sensor and Current Sensor blocks → Simulink scopes/To Workspace.
5. Close the loop: PI controller on ($V_{ref} - V_{out}$), output clamped to [0, 1] as duty.

11.2 Modelling a boost converter

Same block palette; topology per Figure 8: source → inductor → switch to ground; diode from switch node to output capacitor+load. Swap the control direction (duty sets step-up). Simscape Electrical also provides ready-made boost/Buck Converter block libraries (the `Buck Converter` block supports nonsynchronous and synchronous variants) [MathWorks Buck Converter block ref — R17].

11.3 Simulating feedback, duty, load steps, ripple, efficiency

- **Duty sweep**: hold load constant, sweep D from 0.1 → 0.9, plot average V_{out} vs D and compare with $V_{out} = D \cdot V_{in}$ (buck) or $V_{in}/(1-D)$ (boost).
- **Load step**: switch the load resistor (e.g. via an Ideal Switch) at t = 5 ms; observe overshoot/undershoot, settling time; tune PI gains.
- **Ripple study**: zoom steady-state V_{out}; verify ΔI_L against the formula; increase C or L and re-measure.
- **Efficiency**: Simscape lets you enable semiconductor losses (V_{CE}/R_{on}, transition loss models) — the MathWorks thermal-dynamics examples show MOSFET dissipated power averaged over the PWM cycle [R16]. Efficiency $\eta = P_{out}/P_{in}$ from mean values over a long run.

11.4 Testing communication and state transitions

- Model the session as a **Stateflow chart** (or Simulink charts) mirroring Figure 6; drive it with scripted events (plug-in, SLAC done, CableCheck fail, comm timeout).
- Write **test harnesses** that assert: contactors never closed before CableCheck pass; PWM disabled in FAULT within one tick; every state has a timeout path.
- Use **test sequences/assessments** (Simulink Test) and coverage to prove all transitions were exercised.

11.5 HIL at a conceptual level

Hardware-in-the-loop means the *real controller* (your MCU/SECC hardware) runs against a *simulated power plant*: a real-time simulator emulates the converter, battery and EV behaviour and answers the controller's PWM/sensor I/O in real time. You can then safely test fault scenarios (shorted output, sensor drift, EV that violates its limits) that would

destroy hardware on a bench. Companies use HIL to certify protocol conformance (e.g. CharIN test systems) and control software before first power-on. *(Conceptual description.)*

11.6 Common simulation plots and interpretation

Plot	How to read it
V_{out} vs time (startup)	Smooth soft-start, no overshoot past OVP; verify soft-start ramp shape.
I_L waveform	Triangle in CCM; valley hits zero → DCM (loss of regulation); slope = V_L/L.
V_{out} ripple (zoom)	Peak-to-peak vs design target; spikes reveal ESR or layout issues (in simulation: ESR model).
Load-step response	Undershoot depth + settling time; tune PI (too low gain → sluggish; too high → ringing).
Efficiency vs load	Hump curve: low at light load (fixed losses dominate), peak at mid load, droops at heavy load (I^2R).
Duty vs time (fault run)	Saturation at clamp = loop can't reach setpoint → indicates a limit, not a bug.

11.7 Beginner project plan (simulation-first)

1. **Day 1 (a.m.)**: Build open-loop buck (30 V→15 V per MathWorks example); verify D=0.5 → 15 V; capture ripple.
2. **Day 1 (p.m.)**: Close the loop with PI; apply load steps; tune; log settling time.
3. **Day 2 (a.m.)**: Build boost; verify $V_{out} = V_{in}/(1-D)$ at three duty values; add OVP scenario (load disconnect).
4. **Day 2 (p.m.)**: Add thermal/loss models; produce efficiency-vs-load curve; interpret hump.
5. **Day 3**: Stateflow charging state machine (Fig. 6); scripted fault injections (CableCheck fail, comm timeout); produce a one-page report with plots — bring it to the interview as evidence of practical skill.

12. Faults, Protection and Troubleshooting

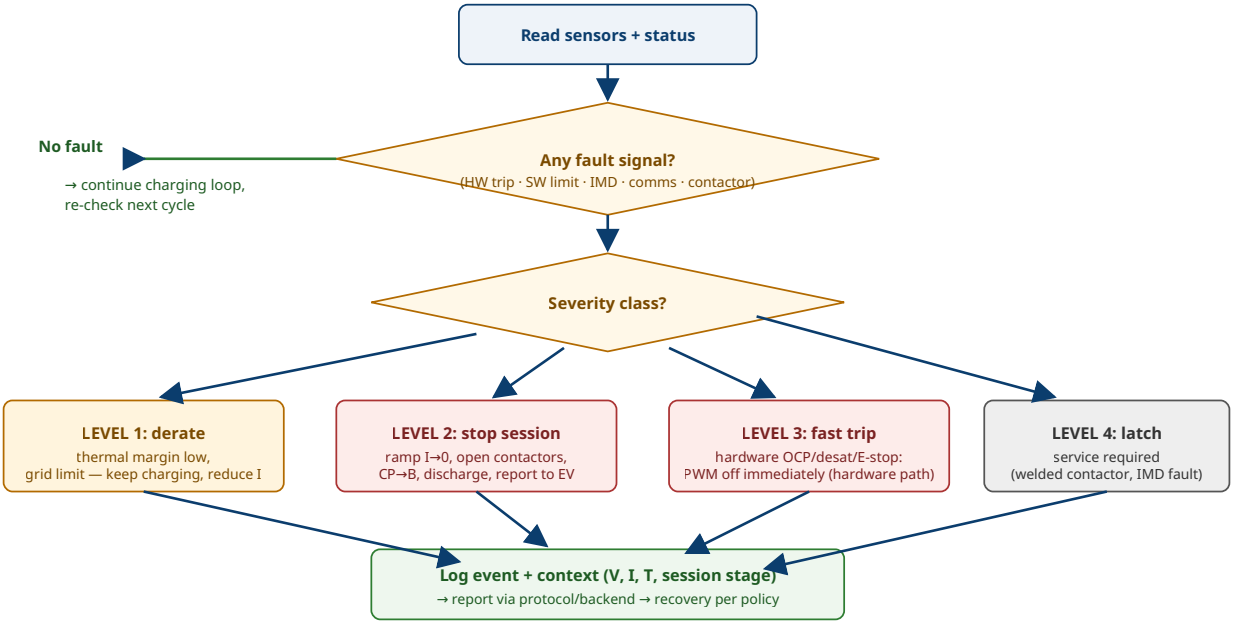
12.1 Fault table

Monitoring/response columns describe typical diagnostic practice; exact certified behaviour is defined by the applicable standard edition and the manufacturer's safety design — do not treat this table as a safety spec.

Fault	Symptom	Likely causes	Monitoring method	Typical safe response
Overvoltage	Output V above limits; EV reports over-voltage; EVSEStatusCode fault	Loop instability, sensor gain error, load dump (EV stops drawing), DC-link runaway	Redundant V sensing (independent channel), software + hardware OVP comparators	Fast PWM disable, open contactors, derate/latch depending on severity; log code
Overcurrent	Current above request/ rating; blown fuse evidence; hot cable	EV drawing beyond agreement, control loop error, sensor offset, short	Shunt/Hall sensors + hardware OCP (desat/ comparator) per switch	Cycle-by-cycle current limit → session stop on repeat; semiconductor fuse / trip
Overtemperature	Module/cable/connector temps high; power derating; fan noise	Cooling failure, ambient heat, loose busbar, blocked vents	NTCs/RTDs on modules, connector (CCS pins), coolant; thermal models	Derate current first; stop session if limits still exceeded
Insulation fault	CableCheck fails; IMD warning/fault; session abort at start or during charge	Damaged cable, moisture/ contamination, EV-side isolation issue, IMD drift	IMD per IEC 61851-23: pre-charge insulation test + continuous monitoring [R10]	Do not start / stop session; keep output discharged; report fault code
Contactor failure	Closing command but no feedback; voltage present when open; session won't start/stop	Welded contacts, coil failure, mechanical jam, driver fault	Aux contacts + pre/post-close voltage signature; welding detection at session end [R5]	Prevent energization; if welded: block session until serviced (hard-latch)
Pre-charge timeout	Output V never matches battery V in the pre-charge window	Battery contactor not closed at EV, output short, DC/DC can't source, wrong request	Pre-charge timer + ΔV and dV/dt monitoring	Abort before main contactor close; open everything; log; retry per policy
Communication timeout	PLC session drops; V2G message not answered in performance time	RFI/noise on CP-PE, EVCC crash, modem firmware bug, loose CP pin	Sequence timers per V2G standard; CP-state watchdog; link-quality stats	Controlled stop of power delivery within standard-defined sequence; log
CP/PLC failure	No State B transition; SLAC never matches; PWM missing	Damaged pilot circuit, diode test fail (State F), injection circuit fault, dirty pins	CP voltage/PWM continuous monitoring, SLAC attenuation stats	No session; display user-friendly error; service code with SLAC logs
Sensor failure	Implausible reading (e.g. −40°C, 0 V while HV present, stuck current)	Broken wire, sensor damage, ADC fault, connector corrosion	Range/rate plausibility + cross-checks between independent sensors	Treat as fault, not truth: fail-safe state; use redundant channel if design allows
DC-link abnormality	Bus V out of band; excessive ripple; bleeder not discharging	Capacitor aging, PFC failure, pre-charge resistor open	Bus V sensor + ripple/ΔV diagnostics; discharge-time check at shutdown	Block session; safe discharge; service flag (cap health trend)
Cooling failure	Pump/fan feedback lost; ΔT across loop too small/large; leak detected	Pump wear, coolant leak, fan bearing, flow sensor failure	Flow switches, pump tach feedback, leak loops, coolant temp sensors	Derate or stop based on thermal margin; prevent cable/module overheating
Emergency stop	E-stop pressed; hard safety chain open	Human action, button/cable fault, interlock open (door)	Hard-wired safety chain monitored independent of main MCU	Immediate hardware-level shutdown (fastest path); latch until reset

12.2 Fault-detection flowchart

Figure 11 — Charger fault-detection and response flow (conceptual)



Conceptual model for learning/diagnostics. Certified designs follow the applicable standard's exact response timings and are validated by type tests.

Figure 11. Severity-graded fault response: derate → stop → hardware trip → service latch, always with logging.

13. Diagram Index and Sources of Visuals

All 13 requested visuals are provided as **original diagrams** created for this report (no copyrighted figures reproduced):

Requested visual	Delivered as	Notes
1	Figure 1 — overall architecture (§2)	Power/control/comms paths colour-coded
2	Figure 13/§3 tables — charger-side hardware	Component map given as the detailed Table 3.1; block-level view inside Figure 1
3	Figure 12 — EV-side block diagram (§4)	Includes "who owns what" split
4	Figure 2 — CCS2 pins (§5.1)	Schematic front view; functions labelled
5	Figure 5 — communication sequence diagram (§5.8)	16 numbered steps, both participants
6	Figure 3 — CP/PWM concept (§5.3)	States + duty-cycle waveform
7	Figure 4 — PLC/SLAC/protocol stack (§5.6)	Layered EV ↔ charger stack
8	Figure 6 — charging state machine (§6)	With fault transitions
9	Figure 7 — buck circuit + ON/OFF paths (§7)	Both switching phases labelled
10	Figure 8 — boost circuit + ON/OFF paths (§8)	Energy storage/release annotated
11	Figure 9 — buck vs boost waveforms (§9)	Gate + inductor current, CCM shapes
12	Figure 10 — MCU/ADC/PWM feedback loop (§10)	Fast/slow loop structure
13	Figure 11 — fault-detection flowchart (§12)	Severity-graded response model

> External figures are not embedded (copyright). Where a standard sequence figure exists publicly (e.g. the ISO 15118-2 DC message-sequence figure in the Typhoon HIL documentation [R5]), it is cited as a reading reference instead, and our own original version is used in this report.

14. Interview Preparation — 30 Q&A

Levels: B = beginner, I = intermediate, S = scenario.

1. How do an EV and a DC charger communicate? B

Two layers: (1) the IEC 61851-1 control pilot — a 1 kHz ±12 V PWM on the CP pin whose voltage levels (states A–F) tell each side whether the vehicle is present, ready and charging; (2) a digital session over power-line communication (HomePlug Green PHY on CP–PE) in which EVCC and SECC exchange DIN 70121/ISO 15118 messages: protocol version agreement, session setup, authentication, charge-parameter exchange, cable check, pre-charge, power delivery and a cyclic current-demand loop. The charger never talks to the BMS directly — it talks to the EVCC.

2. What is the charger controller's role? B

It is the brain of the EVSE: runs the session state machine, converts EV requests into voltage/current setpoints for the power stage via PWM and gate drivers, executes safety functions (insulation check, contactor control, overload/over-temperature protection), metering, user authentication, cooling control and backend (OCPP) communication, and logs everything for diagnostics.

3. What hardware exists on the charger side? B

Power path: breaker/fuses/SPD, EMI filter, PFC rectifier, DC-link capacitors, isolated DC/DC modules with MOSFET/IGBT/SiC switches, gate drivers, output contactors with pre-charge and discharge circuits, CCS2 connector. Control path: MCUs, voltage/current/temperature sensors, ADC/PWM peripherals, IMD, CP/PP circuits, PLC modem/SECC, aux power supplies, cooling, meter, HMI, RFID, connectivity (CAN/Ethernet/cellular), e-stop and interlocks. (See Table 3.1 for functions, signals and failure modes.)

4. What is the difference between CP and PP? B

CP (Control Pilot) is the signalling wire: EVSE-generated PWM whose loaded voltage states (A–F) communicate presence/readiness, plus the PLC carrier in CCS. PP (Proximity Pilot) is a simple resistor-coded wire telling the charger a plug is physically inserted — and in AC charging, the cable's current rating. CP is safety-critical and bidirectional in meaning; PP is a presence/rating signal.

5. What is PLC in CCS2 charging? B

Power Line Communication: data is modulated onto the CP–PE loop using HomePlug Green PHY (≈2–30 MHz OFDM), so no extra data cable is needed. Both ends have a modem (EVCC/SECC) that first runs SLAC to pair, then carries TCP/IP and the V2G protocol messages.

6. What is SLAC? I

Signal Level Attenuation Characterisation — the ISO 15118-3 matching procedure where EV and charger characterize the signal attenuation on the cable to discover each other, exchange CM_SLAC_MATCH parameters, and build a secured logical network before any charging messages flow. It guarantees the right car is talking to the right charger over a shared medium.

7. What are ISO 15118 and DIN 70121? I

DIN 70121 is the earlier, DC-only CCS application protocol — core charging messages, no full PKI security. ISO 15118-2 (2014) extends it to AC+DC, adds TLS and PKI-based Plug & Charge and scheduling; ISO 15118-20 (2022) is the second generation with a new message set and bidirectional power transfer (V2G/BPT). Vehicles negotiate the common version in SupportedAppProtocol.

8. What happens during charging negotiation? I

After session setup and authorization, ChargeParameterDiscovery exchanges limits: EV declares its battery's max/min voltage and current, charger declares its output window; a schedule may be agreed. Then CableCheck verifies insulation, PreCharge matches the charger output to battery voltage, PowerDelivery(Start) energizes, and each CurrentDemand message updates the requested current within the negotiated limits.

9. What is a buck converter? B

A step-down switching regulator: a series switch chops the input, an inductor + capacitor filter it, and the duty cycle D of the PWM sets the average output: $V_{out} = D \cdot V_{in}$ (ideal CCM). The inductor stores energy while the switch is ON and releases it through the freewheeling diode (or synchronous MOSFET) while OFF.

10. What is a boost converter? B

A step-up switching regulator: while the switch is ON the inductor charges from the input; when it opens, the inductor's voltage flips and adds to the input, pushing current through a diode into the output capacitor. Ideal CCM relation: $V_{out} = V_{in}/(1-D)$. Used, for example, as the PFC stage that creates the DC link in chargers.

11. How does PWM control output voltage? B

The switch is ON for a fraction D of each period. The inductor integrates voltage over time (volt-second balance), so the average output scales with D: raising D in a buck raises V; in a boost, raising D shortens the OFF interval, forcing the same inductor current into less time, raising V_{out}. A feedback loop (PI compensator on the sensed output) continuously adjusts D.

12. What is a gate driver? B

A circuit between the MCU's logic PWM and the power switch: it level-shifts and isolates (opto/magnetic/capacitive), sources/sinks amps of gate current to charge/discharge gate capacitance quickly (10–15 V gate rails for MOSFETs, ±15/0 for IGBTs), inserts dead-time in half-bridges, and protects with desaturation/over-current detection that trips the switch faster than software can.

13. Why are voltage and current sensors needed? B

Three reasons: (1) control — the feedback loops need accurate V/I to regulate output; (2) protection — over-current/OVP detection must be fast and independent; (3) compliance/billing — metering and protocol reporting require trustworthy measurements. Redundancy and plausibility checks make the system robust to sensor faults.

14. What is a DC link? B

The intermediate DC bus (with its capacitor bank) between two conversion stages — in a charger, between the PFC rectifier and the isolated DC/DC stage (e.g. ~800 V class, product-dependent). It buffers energy, absorbs switching ripple, and decouples the two stages so each can be controlled semi-independently. It must be pre-charged, monitored and discharged safely.

15. Why are contactors and pre-charge circuits used? I

Contactors provide a physical, galvanic disconnect that survives faults and welding detection, unlike semiconductor switches. Pre-charge avoids closing a contactor into a large ΔV (output capacitors vs battery voltage), which would cause a destructive inrush current/arc: instead the charger ramps its output to battery voltage first (or uses a resistor pre-charge), so the contactor closes at ~0 V difference.

16. What is insulation monitoring? I

Because the DC output floats relative to earth, the IMD measures the insulation resistance between DC+ / DC− and PE using an injected test signal. IEC 61851-23 practice: perform the test before energizing (CableCheck) and monitor continuously during charging; thresholds are commonly expressed in Ω/V of operating voltage — below a defined limit, the session must not start or must stop (summary; verify edition for design values).

17. What happens if communication is lost mid-session? I

The EVSE follows the standard's loss-of-communication procedure: it stops power delivery within defined timeouts, opens contactors, and goes to a safe CP state; the EV similarly stops drawing. State machines treat

message-level timeouts and CP-level loss differently, but both end in safe shutdown, never in "last known setpoint forever".

18. What is a charger state machine? I

The firmware structure that models the session lifecycle (IDLE → PREPARE → SLAC/SESSION → CABLE CHECK → PRECHARGE → CHARGING → RAMPDOWN → DONE, plus FAULT) with guarded transitions, timeouts on every wait, and severity-graded responses. It runs as a hard-real-time task alongside control loops, with hardware interlocks as a backstop (Figure 6, §10.3).

19. How can MATLAB/Simulink help? I

Simulate the power electronics (buck/boost/PFC/DC-DC with real semiconductor models), design and tune the control loops before hardware, model the session state machine in Stateflow and test fault sequences, and run HIL with the real controller against the simulated plant. The MathWorks buck example (30 V→15 V) is a good starting point [R16].

20. How would you debug a charging failure? S

Layer by layer, using logs and measurements instead of guessing: (1) physical layer — plug, CP states, voltages with a scope; (2) PLC layer — SLAC logs, attenuation; (3) protocol layer — V2G message traces, which message failed, EVSEStatusCode; (4) power layer — DC-link present? contactor feedback? insulation test result? Check documentation for error codes, compare a working session, and never bypass protections to force a start.

21. How does embedded C/C++ fit into a charger controller? I

C on the MCU for the hard-real-time parts: ISRs for ADC/PWM, control loops, state machines, drivers, protocol parsers (often MISRA-influenced style, fixed-point where deterministic). C++ (or higher-level code) on application controllers for session management, OCPP, HMI. The discipline: bounded ISRs, no dynamic allocation in critical paths, watchdog coverage, defensive parsing of every external message.

22. Difference between EV-to-charger and charger-to-backend communication? I

EV↔charger = ISO 15118/DIN over PLC, real-time, safety-relevant, controls the power stage, session-scoped. Charger↔backend = OCPP over IP (Ethernet/4G), manages authorization, pricing, telemetry, remote commands; slower, not safety-critical to the power flow, and the charger must behave safely offline. Different stacks, different latency classes, different failure policies.

23. Why is SiC replacing IGBTs in fast chargers? I

SiC MOSFETs switch faster with lower losses at high voltage, allowing higher switching frequency (smaller magnetics), higher efficiency (less cooling), and higher power density — vendors like Infineon target exactly EV-charging power modules with 1200 V SiC families [R13]. Trade-offs: gate-drive sensitivity, cost, EMI design care.

24. What limits how fast a battery charges? B

The BMS decides: cell chemistry and temperature, state of charge (taper at high SOC), cell-voltage headroom, and pack thermal limits. The charger can only deliver what the EV requests; the negotiated CurrentDemand loop tracks these limits. (Charger-side limits also apply: its own output window and thermal capability.)

25. What is OCPP and how does it relate to the session? I

Open Charge Point Protocol — charger↔central-system protocol for authorization, meter values, status, transactions and remote control. It runs in parallel to the EV session: the EV session (ISO 15118) does the power and safety negotiation; OCPP conveys billing/telemetry and can start/stop transactions. They are independent stacks that meet at the charge-transaction record.

26. What does the EVSE do between "plug in" and "power on"? S

CP State B detection and diode check, connector lock, PLC wake and SLAC pairing, IP/TCP(/TLS) setup, protocol negotiation and authorization, ChargeParameterDiscovery limits exchange, CableCheck insulation test (repeats until pass), PreCharge voltage matching, EV closes its contactors, then CP→C and PowerDelivery(Start). Each step has a timeout and a fault path.

27. What is welding detection? I

At session end (and periodically), the system verifies the contactors actually opened and are not welded: e.g. check voltage/current signatures after opening — a welded contactor shows unexpected voltage across it or current through it. ISO 15118-2 defines the WeldingDetection message exchange implementing this per IEC 61851 [R5]. A welded contactor must latch the charger out of service.

28. How does the charger handle an EV requesting less current than expected? S

That's normal operation, not a fault: the BMS tapers current for thermal/SOC/balancing reasons. The charger follows the request within its own limits (its control loop tracks the new setpoint), reports actuals in CurrentDemandRes, and only flags anomalies if the request itself is inconsistent (e.g. below EVMinimumCurrentLimit or contradicting battery data).

29. A sensor reports an implausible value — what should the controller do? S

Trust the plausibility check, not the sensor: cross-check with independent sensors (e.g. second current channel, energy-balance V×I vs meter), check range/rate-of-change limits, and if implausible, fail-safe: derate or stop the session rather than act on bad data; log the event with context. Design-wise, safety functions should use redundant/diverse sensing.

30. Why does the output voltage range matter (e.g. 150–1000 V)? I

Because EVs span 400 V-class and 800 V-class packs, the charger's DC/DC must be able to regulate across that whole window at rated current (or the site stacks modules). Topology choice (wide-gain LLC, DAB, module series/parallel reconfiguration) is driven by this range — a good example of "topology depends on requirements" from §9.3.

15. Practical Scenarios — Reasoning Walkthroughs

Common thread: **documentation → safety → logs → measured evidence → hypothesis → targeted test**. Never bypass a protection to make something "work".

Scenario 1 — The charger detects the EV but charging does not start

Reason it through: Detection (CP State B, SLAC ok) is upstream of authorization/limits. Diagnostic order: 1) session logs — which V2G message failed or timed out? 2) if Authorization failed → check backend link, whitelist, RFID/PnC config. 3) if ChargeParameterDiscovery failed → EV limits vs EVSE output window (e.g. EV wants 800 V, charger max 500 V → no overlap; that's a compatibility issue, not a fault). 4) if CableCheck loops → IMD readings, humidity, connector contamination. 5) if PreCharge times out → EV-side contactor closure confirmed? Output short? Each branch has a distinct log signature.

Scenario 2 — The PLC link cannot be established

Order: 1) confirm CP wiring/injection circuit integrity (SLAC runs on CP–PE). 2) Read SLAC logs: does EV send CM_SLAC_PARM? Does EVSE answer with attenuation profiles? If no frames at all → modem/coupler/cabling. If frames but no match → attenuation too high (long/damaged cable, EMI filter poisoning the band) or firmware/protocol-profile mismatch. 3) Try a second vehicle / a known-good cable to isolate EV vs charger. 4) Check for recent firmware change on either side.

Scenario 3 — Output voltage does not follow the requested value

Order: 1) Safety first: is this a steady offset, oscillation, or saturation at a clamp? 2) Log the setpoint actually received from the session vs what the loop commanded. 3) Check the feedback chain: measure output with a calibrated meter vs the charger's own sensor — if they disagree, sensor/gain calibration issue. 4) If sensor is right: loop instability (scope the control PWM, look for hunting), or the module hit an internal limit (DC-link sag, thermal derate). 5) Verify the DC-link voltage supports the requested output (output ≤ link − margin for buck-derived stages).

Scenario 4 — Current readings are unstable/noisy

Order: 1) Compare two independent current measurements (sensor vs meter) — if both noisy, real ripple; if only the internal one, sensor/ADC issue. 2) Check ADC sampling synchronization with PWM (aliasing of switching ripple into the readings). 3) Inspect grounding/shielding of the sensor cable (EMI pickup). 4) Check load side: is the EV itself oscillating (some vehicles hunt with poorly damped loops)? 5) Look at control loop gains — recently changed? Oscillation frequency vs loop bandwidth.

Scenario 5 — The converter overheats

Order: 1) Confirm with IR camera — which component: switch, inductor, busbar? 2) Check cooling first (fans/pumps, filters, coolant flow, ambient temperature) — most common cause. 3) Review the duty cycle/power profile at the time (derating curves respected?). 4) If a specific switch is hot → gate-drive issue (slow switching), loose termination, or degraded device (compare $R_{DS(on)}/V_{CE}$ trend across modules). 5) Verify thermal derating logic actually reduced current — if it didn't, that's a firmware/logic bug to file.

Scenario 6 — The contactor does not close

Order: 1) Is a close command even issued? (State machine stuck upstream — check why: safety chain open, pre-charge incomplete.) 2) Coil voltage present at the contactor? If yes but no click → mechanical/coil fault. 3) Aux contact feedback vs actual state — miswired or broken aux. 4) If it closed once and never again → possible welding latch from a previous session (check the welding-detection logs and output-voltage-present flag). 5) Inspect interlock chain (e-stop, doors) before blaming the contactor.

Scenario 7 — Pre-charge does not complete

Order: 1) Confirm from logs: output voltage ramped but never matched the battery? or didn't ramp at all? 2) Didn't ramp → DC/DC setpoint chain, or output short (then IMD might also flag). 3) Ramped but ΔV stays large → EV-side contactor not actually closed (ask the EV: its log/status), or EV mis-reporting its battery voltage. 4) Timeout too aggressive for a very large EV pack? Compare configured timeout vs pack size. 5) Check pre-charge resistor health if it's a resistor-based design (thermal/continuity).

Scenario 8 — Charging stops unexpectedly mid-session

Order: 1) Pull the session log: who stopped it — EV (PowerDelivery Stop with reason), EVSE fault, or backend? 2) If EVSE fault → which code, at what V/I/T operating point? Pattern (e.g. always at same current → thermal; random → comms). 3) If comm timeout → PLC quality stats, EMI events (e.g. compressor start nearby), CP pin integrity. 4) If EV stopped → the reason code from the vehicle (SOC reached, battery protection). 5) Reproduce with a controlled load/EV simulator if possible — real evidence beats speculation.

Scenario 9 — The EV requests less current than expected

Order: 1) This is usually correct behaviour: BMS tapers on SOC/temperature/cell-voltage. Check the trend: smooth taper = normal; sudden drop = battery protection event. 2) Compare against a session matrix for that EV model (some vehicles simply cap below the charger's rating — compatibility, not fault). 3) If the request contradicts battery data (e.g. cold battery full current) → firmware/policy question with the OEM. 4) Confirm the charger is not the one limiting (derating active? grid limit?).

Scenario 10 — A sensor reports an implausible value

Order: 1) Plausibility rules should already have flagged it (range, rate, cross-sensor). 2) Verify physically at the connector/sensor with a meter — wiring/connector first (most common). 3) Check calibration constants/configuration for that channel (wrong scaling after an update?). 4) Check whether the ADC channel itself reads other sensors correctly (peripheral vs sensor). 5) File the event with full context; if safety-relevant, fail-safe now and diagnose after.

Scenario 11 (bonus) — Random trips only at certain sites

Order: 1) Compare site electrical environment: grid quality (sags/surges), grounding quality, nearby interference. 2) Check whether trips correlate with grid events via input sensing logs. 3) Review SPD counter and input filter health. 4) Site-specific EMI (e.g. large VFDs nearby) affecting PLC quality → SLAC/comm timeout pattern. 5) Produce a site report with measured evidence — don't tune firmware to mask a site problem.

Scenario 12 (bonus) — New EV model charges, but slowly, on all your chargers

Order: 1) Capture a full V2G trace: what does ChargeParameterDiscovery say the EV's limits are? 2) Check the EV's request curve over SOC — normal taper or flat low cap? 3) Compare EV-side vs charger-side declared limits — narrow EV max current/voltage = vehicle design choice. 4) Document as a compatibility profile (charging curve) — this is exactly how real interoperability teams work.

16. Glossary, Standards Map and 3-Day Study Roadmap

16.1 Glossary of abbreviations

Abbreviation	Full term	Meaning in one line
AC / DC	Alternating / Direct Current	Grid vs battery-side electrical domains
BMS	Battery Management System	Vehicle system managing cell safety and charge limits
CCS	Combined Charging System	Charging interface family (CCS1/CCS2) combining AC pins + DC pins
CP / PP	Control Pilot / Proximity Pilot	Signalling pin (PWM+PLC) / plug-presence pin
DCFC	DC Fast Charger	Off-board DC charging station
DAB	Dual Active Bridge	Isolated bidirectional DC/DC topology
EMI / EMC	Electromagnetic Interference / Compatibility	Noise suppression and immunity design
EVCC / SECC	EV / Supply-Equipment Communication Controller	The two PLC session endpoints
EVSE	Electric Vehicle Supply Equipment	Standards term for the charging station
HPGP	HomePlug Green PHY	PLC physical layer used by ISO 15118-3
HIL	Hardware-in-the-Loop	Real controller tested against simulated plant
IMD	Insulation Monitoring Device	Measures DC-output-to-PE insulation resistance
LLC	Inductor-Inductor-Capacitor resonant converter	Common isolated DC/DC topology in chargers
MCU	Microcontroller Unit	Embedded processor running control/firmware
OBC	On-Board Charger	In-car AC charger (bypassed in DC charging)
OCPP	Open Charge Point Protocol	Charger↔backend management protocol
OVP / OCP	Over-Voltage / Over-Current Protection	Protection functions (hardware and software)
PFC	Power Factor Correction	Front-end stage shaping grid current, making the DC link
PKI	Public Key Infrastructure	Certificate system behind Plug & Charge
PLC	Power Line Communication	Data over the CP–PE power/signalling lines
PWM	Pulse-Width Modulation	Duty-cycle-controlled switching
SLAC	Signal Level Attenuation Characterisation	EV↔EVSE pairing procedure over PLC
SiC	Silicon Carbide	Wide-bandgap semiconductor for HV switches
SPD	Surge Protection Device	Clamps lightning/grid surges
TLS	Transport Layer Security	Encryption layer used by ISO 15118-2 PnC
VCU	Vehicle Control Unit	Vehicle-level coordination controller
V2G	Vehicle-to-Grid	Bidirectional energy exchange concept/standard family
CCM / DCM	Continuous / Discontinuous Conduction Mode	Inductor current never / sometimes reaches zero

16.2 3-day focused study plan (prioritised to the manager's topics)

When	Focus	What to do (concrete)
Day 1 morning	Big picture + architecture	Read §1–§2; redraw Figure 1 from memory (power/control/comms paths); say aloud what each block does. Explain AC vs DC charging in 60 seconds.

Day 1 afternoon	Hardware	Skim Table 3.1 twice; for 10 key components (breaker→contactor), rehearse: function, failure mode, how tested. Memorise the "power stage vs control stage" split (§1.4).
Day 2 morning	Communication (the manager's #1 topic)	Read §5 fully; redraw Figure 3 (CP states) and Figure 5 (sequence) from memory. Practise saying the 17-step session story (§6) in 2 minutes and in 10 minutes. Memorise the message names and what each does (Table in §5.7).
Day 2 afternoon	Buck/boost	Read §7–§9; redo both worked examples on paper without looking; derive $V_{out}=D\cdot V_{in}$ and $V_{out}=V_{in}/(1-D)$ via volt-second balance; practise Figure 9 interpretation.
Day 3 morning	Embedded + simulation	Read §10–§11; write the pseudocode state machine once by hand; if MATLAB is available, build the 30 V→15 V buck per the MathWorks example and screenshot the response — bring it as a portfolio artifact.
Day 3 afternoon	Faults + interview drill	Read §12 and §15; do all 30 Q&A (§14) out loud; pick 3 scenarios and explain diagnostic steps to a friend. Prepare 3 questions to ask the manager (their topology? protocol versions? HIL setup?).

> **Interview mindset:** when unsure, reason from first principles (what does safety require? what does the energy path need? what does the feedback loop see?) and say "that will be architecture/standard-dependent, but here's my reasoning…". Managers hire for reasoning, not recall.

17. References

All URLs accessed and verified on **9 October 2026**. Where a claim is a summary of a standard by a third party, the citation says so; consult the actual standards (IEC 61851-1/-23, ISO 15118-2/-3/-20, DIN 70121) for normative requirements.

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> **Verification note.** Protocol message names and sequence order were verified against [R5]/[R18]; CP states and levels against [R1]/[R2] (secondary summaries); buck/boost equations against [R3] and standard theory; insulation practice against [R10]/[R11]; Simulink workflow against [R16]/[R17]. Items flagged *architecture-dependent* or *illustrative* in the text are not verifiable universal facts and should be checked against your target company's actual products and the current standard editions.